How Much Does It Take to Top a Multi-Task Leaderboard? Minimum Targeted Investment under Different Aggregation Rules

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Abstract

A developer can try to put one model first on a multi-task leaderboard by improving it on some of the sub-datasets. How many score points does that take, and does the answer depend on how the leaderboard aggregates? We define investment as points added on chosen sub-datasets with all other models fixed and each score capped, measure it three ways (tasks touched, total points, largest single gain; each minimised exactly), and solve it for six score rules and three rank rules on every non-leading model of HELM MMLU (78 targets, 57 tasks) and the Open LLM Leaderboard (4495 targets, 6 tasks). The answer depends on the rule, the cost measure and above all the per-task cap. With no cap the median total cost on HELM is 696 points under the mean and 112 under the median; on HELM the median and minimum rules are five to ten times cheaper than the mean in total points, on the OLLB census only the minimum is several times cheaper (about two times after normalising by each rule's score scale), and in the largest single gain none is. The smallest uniform gain per task at which half of the targets can reach first place is 6–22 points on HELM and 18–36 on OLLB depending on the rule, in a different order on the two leaderboards. With 10 points per task only 3% (mean) of OLLB targets can reach first place. Raising every task by the nominal minimum puts a target first in a median of only 0.14–0.43 of noise draws on OLLB and 0.14–0.80 on HELM (36 targets per dataset, noise with the published sampling error); reaching 90% takes a further 1.3–2.5 points per task, and more for the minimum rule. The caps are illustrative and not calibrated to what data buy in points. Results hold for these two leaderboards, one deviating developer, fixed rivals and costs in leaderboard points.

1 Introduction

A multi-task leaderboard is a matrix of scores, one row per model and one column per sub-dataset, plus an aggregation rule that turns each row into a rank. Rank affects adoption, publication and funding, so developers have a reason to act on it. One available action is to add data related to some sub-datasets to the training of a particular model, which raises its scores on those sub-datasets (Zhou et al., 2023; Yang et al., 2023). How much of this does a developer need to put a chosen model first, and does the answer depend on whether the leaderboard averages, takes a median, or counts wins?

We study this as a minimum-cost problem on public score matrices. Our setting is deliberately simple. One target model may raise its score on any set of tasks, up to a cap. Every other model is fixed. The target is first if it is strictly ahead of every rival under the rule. The cost is measured in leaderboard points, three ways: tasks touched (K), total points added (L_1) and largest single-task gain (L_∞), each minimised separately and exactly. The cap is the main modelling assumption, so it is a swept parameter, from the theoretical ceiling to fixed per-task gains and empirical gains of fine-tuned models over their base models.

Contributions. (i) A formal model of targeted investment under an aggregation rule, with three cost measures, a cap parameter, and exact solvers for six score rules and three rank rules (Section 3). The closed forms for the mean, minimum and median are elementary; we state them to validate the solvers and to say what the rules cost in the simplest case (Appendix B). What is not

derivable in a few lines is the exact treatment of the trimmed mean and the rank rules, by mixed-integer programmes, and the results below. (ii) An evaluation on every non-leading model of two public leaderboards: HELM MMLU (Liang et al., 2022; Hendrycks et al., 2021) and the Open LLM Leaderboard, with uncertainty over targets, over evaluation noise, over task subsets and over rule-specific choices (Section 4). The empirical contribution is the map of how cost and feasibility depend on rule, cost measure and cap, and how a fixed investment fares under evaluation noise. (iii) Six claims fixed in the plan before the grids, reported with criterion, measured value and verdict, including where a criterion was loose or a claim reworded after the results (Table 1).

Findings. *Cost depends on the rule, but not in a single order.* In total points the mean rule is several times dearer than the minimum rule on both leaderboards and the median rule on HELM, but not in the largest single gain, and the number of tasks it needs is among the smallest (Section 4.1). *Caps decide whether first place is reachable at all.* With 10 points per task at most 10% of OLLB targets can reach first place under any score rule; the cap at which half of the targets can win is 6–22 points on HELM and 18–36 on OLLB, in a different rule order on each (Section 4.2). *Task-count intuitions do not generalise*: $K(\text{mean}) \leq K(\text{trimmed}) \leq K(\text{median}) \leq K(\text{min})$ fails for many targets (Section 4.3). *Rank rules are cheaper in points than the mean when unconstrained; under caps plurality and the Condorcet requirement are feasible for fewer targets and Borda for more* (Section 4.4). *Near the top the nominal minimum is not enough under noise* (Section 4.5).

What we do not claim. Investment is measured in score points on the leaderboard, not in training data, compute or money: we do not model how many points a given amount of data buys. We study one deviating developer against fixed rivals, on two leaderboards which are both narrow (Section 5).

2 Related work

Benchmarks and leaderboards. MMLU (Hendrycks et al., 2021) gathers 57 subjects, HELM (Liang et al., 2022) reports per-subject scores for many models, and the Open LLM Leaderboard ranks models by the mean of six task scores computed with the lm-evaluation-harness (Gao et al., 2023). These are the leaderboards we use. A different choice of tasks can reorder methods (Dehghani et al., 2021); we ask the reverse question, how much one model must change under fixed tasks to top the board.

Aggregation as social choice. Several works treat tasks as voters and models as candidates, and propose voting rules to aggregate benchmark results (Lanctot et al., 2023; Rofin et al., 2023); the Handbook of Computational Social Choice (Brandt et al., 2016) covers the rules we use (Borda, plurality, Condorcet-consistent rules). The theory of bribery and of the margin of victory asks how many voters must change their votes to change a winner (Faliszewski et al., 2009; Xia, 2012). Our problem is the score-valued analogue: the voters are tasks, a change is a raise of one score, and the cost is the points raised. We differ in using real leaderboard matrices, continuous capped score changes and cost measures that suit them. We claim no reduction to, and no complexity result for, bribery or margin of victory: hardness for discrete bribery does not transfer automatically to capped real-valued changes, and for rank rules we rely on a mixed-integer solver that proved optimality or infeasibility on every run reported here.

Gaming and robustness of rankings. Test-set leakage and rephrased test data inflate scores (Zhou et al., 2023; Yang et al., 2023); Singh et al. (2025) show how private testing and uneven sampling shape the Chatbot Arena leaderboard; Huang et al. (2025) show that dropping a very small fraction of preference data can change the top-ranked model in Bradley–Terry rankings. We take the mechanism as given and ask what a score-level intervention costs under each rule.

Strategic developers. Chen et al. (2026) model benchmarking as a Stackelberg game between a benchmark designer and several developers, and prove that current benchmarks induce games with no Nash equilibrium among developers, while tune-before-test gives a unique equilibrium. Ours is a single deviator against fixed rivals on public score matrices, the minimal-cost counterpart and first move of such a game; it shares no theorem with theirs.

3 PROBLEM AND METHOD

Setting. A leaderboard is a score matrix $X \in [0, 100]^{M \times n}$ of M models on n tasks. A rule f maps each score vector to a value (score rules) or each matrix to an outcome (rank rules), and ranks models by it. Fix a target model i with scores $x = X_{i\cdot}$. An *intervention* is a vector y with $x_t \le y_t \le c_t$, where c_t is a cap on task t. All other models stay fixed: no rival responds and no spillover moves other tasks. The target is *first* if it is strictly ahead of every rival under f by a margin $\varepsilon = 10^{-3}$ points (for rank rules, strictly better outcome on a discrete level model in which a tie with a rival counts half).

Investment. The cost of y is measured in leaderboard points, in three ways: $K = |\{t : y_t > x_t\}|$ (tasks touched), $L_1 = \sum_t (y_t - x_t)$ (total points) and $L_\infty = \max_t (y_t - x_t)$ (largest single-task gain; the uniform "general improvement" benchmark). Each is the exact minimum of its own problem, so the minimiser of one is generally not that of another and they are never plotted as one strategy. *This is not a cost in data, compute or money.* The map from a data effort to points on a task is outside the model; the cap is where we encode what is plausible.

Caps. The cap is the main modelling assumption, so it is a swept parameter: ceil (100, a theoretical upper bound only), front0/front2 (the best rival's score on that task plus 0 or 2 points), gain5/gain10/gain20 (at most that many points per task), and emp95/emp99 (per-task 95th/99th percentiles of fine-tune-minus-base differences observed on OLLB; 540 pairs). The empirical caps reflect general fine-tuning gains on six tasks, not the effect of task-specific data; on HELM only the other caps are used. An infeasible target (no raise within the cap reaches first place) is recorded as such and never dropped.

Rules. Score rules: arithmetic mean (the OLLB rule), geometric and harmonic mean (on scores shifted by 1, checked with other offsets), median, minimum, trimmed mean (drop the best and worst task; OLLB only), and weighted means as a sensitivity check. Rank rules: Borda, plurality (number of tasks on which a model is best), and the Condorcet requirement (the target must beat every rival on a majority of tasks). The last stands for Condorcet-consistent rules (Copeland, Schulze and others), not for Borda or Kemeny, and Kemeny is out of scope. Score rules differ in which part of the score vector matters (all of it, the middle, the minimum); rank rules ignore magnitude.

Exact solvers. For the mean the minimum has a closed form: with deficit $R = nb - \sum_t x_t$ and b the best rival's value plus ε, $L_1 = R$ whenever the headroom suffices, K is the least number of the largest headrooms that cover R, and L_∞ is a water level. The minimum and median also have closed forms for K, L_1 and L_∞ (odd n; for even n only K), and none of them is "n tasks" or "$\lceil n/2 \rceil + 1$ tasks" in general, because tasks already above the winning level count (Appendix B). Geometric and power means are solved by water-filling with bisection, the trimmed mean and rank rules by mixed-integer programmes solved with HiGHS (Huangfu & Hall, 2018) with lazily added rival constraints. Every solver is tested against brute force on small random instances, and the closed forms against solver output on every target at the ceiling cap: 0 mismatches in 16 cells (largest difference 9.4e-08); a numerical agreement, not a proof.

Targets and denominators. The targets are every model except the leader under the plain arithmetic mean: 78 of 79 on HELM and 4495 of 4496 on OLLB for score rules. Rank rules on OLLB are costly, so they are run on a rank-stratified sample of 136 targets, a deterministic grid (ranks 2–21 plus log-spaced ranks), not a population sample; it is labelled wherever it appears. A target already first under a non-mean rule has zero cost under it and stays in the comparison. Costs are summarised as quantiles over targets and, for capped runs, over feasible targets only, together with the feasible share (Wilson 95% interval).

Uncertainty. We keep three kinds of interval apart. *Wilson intervals* describe a share of targets. *Bootstrap over targets* (2000 resamples, percentile) describes the sampling variability of a median over the target models; it is not evaluation noise. *Draw quantiles* describe the spread over replicates for one target: evaluation noise (binomial standard errors per cell from the item counts), bootstrap over tasks, random task subsets, and random weights; these are the quantity of interest and not confidence intervals. Evaluation-noise draws give a distribution of the re-optimised minimum cost.

Table 1: Pre-stated claims: criterion fixed in the plan, measured value, verdict.

	Criterion in the plan	Measured	Verdict
C1	Closed forms (mean, minimum, median) match the solvers.	0 mismatches in 16 cells (largest difference 9.4e-08).	Holds; a solver check.
C2	K(mean)$\leq$ K(trimmed)$\leq$ K(median)$\leq$ K(min); refuted if the order fails "for a sizeable share of targets" (no number fixed).	K(median)$\leq$ K(min) at ceil for HELM 9%, OLLB 89% of targets; OLLB at gain10, the three steps hold for 36, 6, 29%.	Refuted as a general order (post hoc: a step that fails for over 10% of targets).
C3	Almost every rule gameable at ceil; "sharply" different feasibility under caps; refuted if rule order and feasibility are stable across caps.	Feasible for 100% of targets at ceil, front0, front2. At gain10: OLLB 1–10% by score rule, HELM 18–73% by rule.	Supported.
C4	Rank-rule cost is not a monotone function of the score gap; "reported either way"; no criterion.	At ceil Borda, the Condorcet requirement and plurality are cheaper in L_1 than the mean for 99, 100 and 100% of HELM targets; under caps they are infeasible more often (Section 4.4).	Holds at ceil; qualified by feasibility.
C5	Plan wording: cost for top targets is smaller than evaluation noise; first-place probability against investment to be reported. No threshold.	Post hoc restatement: a uniform raise by the nominal minimum L_∞ puts the target first in a median of 0.14–0.43 (OLLB) and 0.14–0.80 (HELM) of noise draws over 36 targets per dataset; 90% needs a further 1.3–2.5 points (all rules but minimum).	Plan wording not tested as written; restatement supported.
C6	K/n and L_1 against n, with and without cross-task correlation; no criterion.	Mean L_1 at $n = 3$ and $n = 57$: 6.1 and 154.0 observed, 49.4 shuffled at $n = 57$.	Descriptive; no claim that could fail.

For "how likely is a fixed investment to put the target first" we replay a strategy chosen on the observed leaderboard (a uniform raise of d points on every task up to the cap, or the same L_1 on the tasks with most headroom) against noise draws: each cell gets an independent Gaussian error with the binomial standard error of that cell (item counts, observed score), clipped to $[0, 100]$; the success curves depend on this assumption. Wilson bands on replay curves are pointwise Monte Carlo intervals over the draws, and target-bootstrap bars resample the observed target models, not evaluation noise and not a population of models.

4 Experiments

Data. *HELM MMLU*, release v1.13.0 (summary date 2025-01-10): 79 models on 57 subjects, exact-match accuracy in points, no missing cells. *Open LLM Leaderboard* (OLLB), the `contents` dataset at Hugging Face commit `9c09a7c` (2025-03-20; parquet file SHA-256 in Appendix A): $4495 + 1$ models on six tasks (IFEval, BBH, MATH Lvl 5, GPQA, MuSR, MMLU-Pro) in the leaderboard's normalised points, after removing 79 duplicate names (the best variant is kept; the variant keeping the worst is a robustness slice) and one flagged model. The OLLB rule is the unweighted mean of the six normalised scores, which we verified reproduces its published average. We use only scores; neither source declares a licence for them. All numbers below are regenerated by the scripts in Appendix A.

Pre-stated claims. Before running the grids the plan fixed six claims. Table 1 gives, for each, the refutation criterion as it was written, the measured value and the verdict. Where the plan fixed no number (C2, C3, C4, C6) any threshold we apply in the verdict was set after seeing the results (post hoc); C5 was reworded after the results, and marked as such.

4.1 Headline: cost by rule at the ceiling cap

Table 2 gives the median over targets of the three costs at the ceiling cap. Figure 1 shows the costs against the target's current mean rank under the front0 cap, which is feasible for every target and binding, on the 136 OLLB targets on which all nine rules were run: L_1 rises with the original rank under every rule, fastest for the mean and slowest for the minimum, and K stays between 1 and 6 tasks. On OLLB the mean needs the *fewest* tasks (median $K = 2$ of 6) and the *most* points

Table 2: Ceiling-cap costs: median [25th, 75th percentile] over feasible mean nonleaders. OLLB uses its 4495-target census; HELM uses 78 targets. Feasibility is explicit; the three metrics have separate optima. Points are task score increases, not data or compute budgets.

Rule	Feasible	K	L_1	L_∞
OLLB census, ceiling cap				
Mean	4495/4495	2.0 [2.0, 3.0]	180.5 [136.7, 228.8]	30.1 [22.8, 38.1]
Geometric	4495/4495	3.0 [2.0, 4.0]	153.2 [113.2, 200.8]	27.8 [21.7, 34.8]
Harmonic	4495/4495	3.0 [2.0, 4.0]	124.8 [90.0, 171.3]	24.9 [19.7, 31.0]
Trimmed	4495/4495	3.0 [2.0, 3.0]	141.1 [110.7, 184.2]	33.7 [27.0, 41.4]
Median	4495/4495	3.0 [3.0, 3.0]	105.9 [78.5, 145.8]	36.2 [28.5, 44.4]
Minimum	4495/4495	3.0 [3.0, 5.0]	43.2 [29.5, 67.7]	17.9 [14.6, 21.5]
HELM, ceiling cap				
Mean	78/78	16.0 [10.0, 22.8]	696.5 [443.3, 1056.3]	13.2 [8.2, 20.3]
Geometric	78/78	16.0 [10.2, 22.8]	668.7 [422.3, 1027.6]	14.2 [9.0, 21.8]
Harmonic	78/78	16.0 [10.2, 22.8]	643.6 [404.3, 1000.6]	15.3 [10.0, 23.7]
Median	78/78	21.0 [14.0, 27.0]	112.3 [40.4, 253.2]	9.8 [5.8, 16.3]
Minimum	78/78	11.5 [7.0, 18.0]	98.1 [46.1, 240.6]	22.2 [13.4, 28.9]

($L_1 = 181$), against $L_1 = 106$ for the median and 43 for the minimum; its L_1 exceeds the median's for all but 0.1% of targets and the minimum's for all but 0.0%. On HELM (57 tasks) the mean needs $K = 16$ and $L_1 = 696$, the median $K = 21$ and $L_1 = 112$, the minimum $K = 11.5$ and $L_1 = 98$. The cheapest rule depends on the measure: the minimum has the smallest L_∞ on OLLB (17.9) but the largest on HELM (22.2), where it must lift one weak subject far above the others. The mean is dear in points because $L_1 = n(\text{gap} + \varepsilon)$, the whole deficit summed over n tasks, whereas the median and minimum repair only the tasks that matter to them (Appendix B).

Is the gap between rules an artefact of score scale? The rules' scores live on different scales, so we divide each rule's cost by its own leader gap or by the interquartile range of its score, and also match targets on rank under the rule (Appendix D). The ratio of the median L_1 to the mean's is on HELM 0.16 (median rule) and 0.14 (minimum) raw, and between 0.09 and 0.19 in all normalised and rank-matched views: five- to ten-fold. On the OLLB sample both rules are about three-fold cheaper (raw 0.36, 0.29), but the minimum's ratio rises to 0.66 by IQR. On the OLLB census only the minimum is several-fold cheaper (raw 0.24, normalised 0.41–0.54); the median rule's ratio is 0.59 raw and 0.50–0.57 normalised. The advantage is one of L_1 only: the ratio of median L_∞ to the mean's is 0.74 (median rule) and 1.69 (minimum) on HELM and 1.20 and 0.59 on the census. Models with a zero on some task (7.0% of OLLB) do not explain the minimum's low cost: removing them gives the ratio 0.235 (0.239 with them).

4.2 Caps decide feasibility

Figure 3 is the cap map. At ceil, front0 and front2 every target is feasible for every rule, by construction or because the ceiling is 100. Under per-task gain caps the picture changes. With gains of at most 10 points per task, the share of OLLB targets that can reach first place is 3% under the mean and between 1% (trimmed mean) and 10% (minimum) over all score rules, and on HELM it ranges from 18% (minimum) to 73% (plurality) over all rules. With gains of at most 20 points the OLLB shares are 18% (mean) up to 64% (minimum). With the empirical caps they are 7% (mean, emp95) and 12% (mean, emp99). Costs at the ceiling are therefore no guide to realistic cost; under a cap a rule can look cheap only because it is feasible for the few targets already close, so medians are over feasible targets with the feasible share. The cap also changes the cost ordering. On HELM the number of tasks of the mean is 16 at ceil but 37 at front0, while its L_1 is unchanged (696 and 696), because the points needed are the same and only their spread over tasks is limited by the rivals' scores. Among the eight rules, the rank of the mean by median K falls from 3 (tied) at ceil to 7 at front0. C3 is supported: feasibility and ordering both depend on the cap.

A threshold instead of a grid. The smallest uniform cap at which a target can reach first place is its L_∞ at the ceiling (feasibility is monotone in the cap; this agrees with the stored 5-, 10- and 20-point solves in all 84,006 (target, rule, cap) cells, 0 disagreements). Call it g^*; the share of

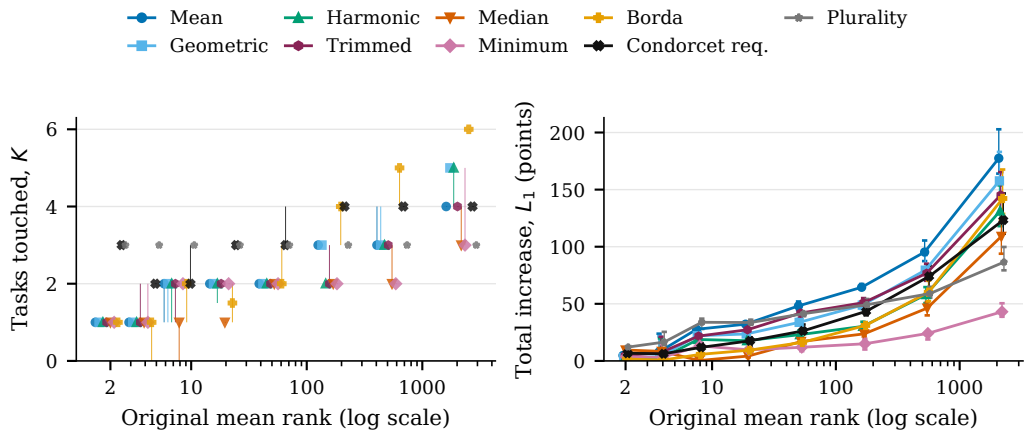


Figure 1: Costs under the front0 cap on the same 136 OLLB sampled targets (six tasks). Points are rank-bin medians; whiskers are 95% target-bootstrap intervals. K and L_1 are separately minimized. All shown targets are feasible. Rule offsets separate overlapping points; the K panel has no connecting lines.

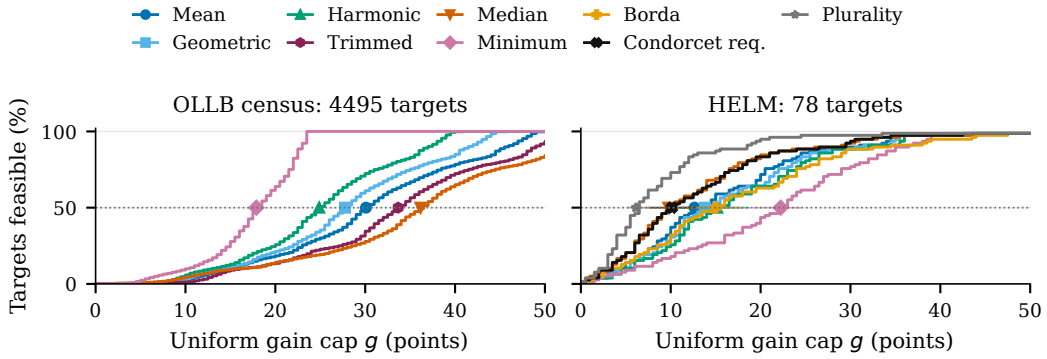


Figure 2: Feasible shares against the uniform gain cap g: OLLB census (4495 targets) and HELM (78). Markers give 50% crossings; bars show 95% target-bootstrap intervals.

feasible targets at cap g is the empirical distribution function of g^* (Figure 2). The cap at which half of the targets can win ranges, over rules, from 6 to 22 points on HELM (mean 12.6, 95% interval 10.2–16.4) and from 18 to 36 on the OLLB census (mean 30.1, 29.7–30.4). The order of rules by this threshold differs between the leaderboards: the median rule is among the cheapest on HELM (9.8) and the dearest on the OLLB census (36.2), the minimum the dearest on HELM (22.2) and the cheapest on the census (17.9), and the order by g^* differs from the order by L_1. For scale, among 540 observed pairs of a fine-tune and its base model on the leaderboard, the 99th percentile of the gain is below the census median g^* of the mean on 5 of the six tasks (at most 26 points; IFEval 63); the pairs are a small selected subset, so this anchors the scale of g^* and does not model what investment can buy.

Dependence between models. OLLB models of one family are not independent targets. Resampling the 3849 base-model families that contain the 4495 targets (3744 singletons) widens the intervals by a median factor of 1.7 (range 1.0–2.7); the mean's median L_1 at the ceiling is 180.5 with cluster interval 176.0–184.4 against 177.9–182.3 over targets. Dropping families of more than 10, 20 or 50 members changes no median L_1 by more than 1.9% and no feasible share by more than 0.02. HELM has no family labels.

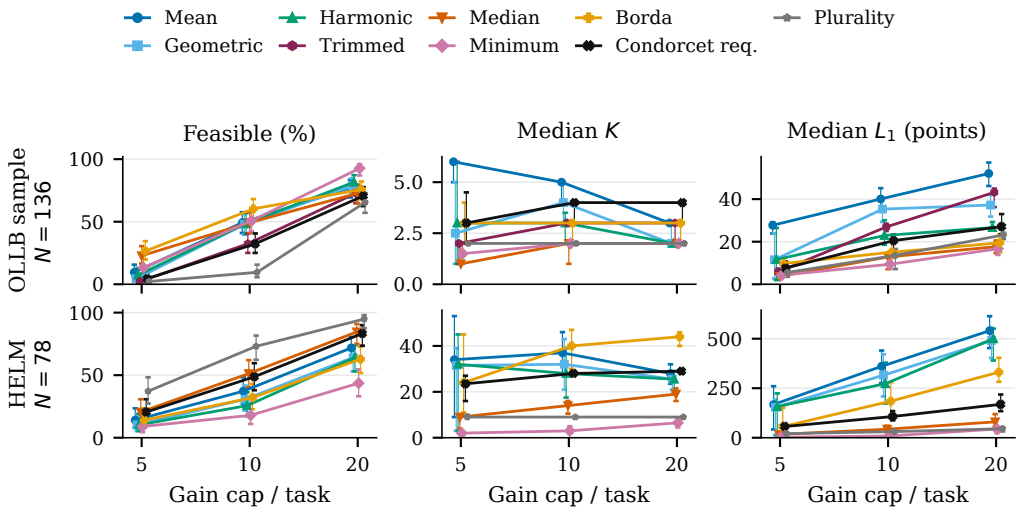


Figure 3: Binding gain caps change feasibility and conditional costs. OLLB uses the matched 136-target sample; HELM uses all 78 mean nonleaders. Shares have Wilson 95% intervals; feasible-target cost medians have 95% bootstrap intervals. Different feasible cohorts underlie the cost curves.

4.3 Task-count orderings do not generalise

C2 formalises a task-count intuition from our own plan, not from a published statement: the more of the score vector a rule uses, the fewer tasks it needs, $K(\text{mean}) \leq K(\text{trimmed}) \leq K(\text{median}) \leq K(\text{min})$. We use it as a test case of why task counts mislead. On OLLB at ceil the first two inequalities hold for 100% and 100% of the 4495 targets, and the last for 89%. At front0 the three shares drop to 74%, 61% and 82%, and on HELM at ceil the mean-median and median-minimum inequalities hold for 86% and 9% of targets. We treat C2 as refuted as a general order. Each rule has its own rival threshold, so the number of tasks to raise is not ordered by how strict a rule seems; likewise "the median needs $\lceil n/2 \rceil + 1$ tasks" and "the minimum needs all n" are wrong in general, since tasks already above the winning level count (Appendix B).

4.4 Rank rules against the mean

Rank rules ignore magnitude, so their cost is not a function of the score gap (C4). At the ceiling cap they are cheaper in points than the mean for nearly all targets: on HELM the median ratio of L_1 (rank rule over mean) is 0.78 for Borda, 0.30 for the Condorcet requirement and 0.07 for plurality, cheaper for 99%, 100% and 100% of targets. On the OLLB sample (136 targets, rank-stratified) the ratios are 0.46, 0.66 and 0.42. The picture reverses in feasibility. Under 10-point gains on the OLLB sample, plurality is infeasible for 54 targets for which the mean is feasible, and feasible for 0 where the mean is not; Condorcet is infeasible for 24 such targets. Borda, in contrast, is feasible for 17 targets where the mean is not. Whether a rank rule is the cheaper route therefore depends on the cap and on the rule. The OLLB sample is a deterministic grid of ranks, not a random sample, so the sampling step carries no seed. Redrawing the tail ranks with three random seeds (126–136 targets each, ranks 2–21 always included) leaves the median L_1 ratio at 0.46–0.49 for Borda, 0.41–0.42 for plurality and 0.65–0.66 for the Condorcet requirement; the rank rule is cheaper for at least 98.4% of targets; and at gain10 the number of targets where plurality is infeasible but the mean feasible stays at 41–54 (Condorcet 19–25); every redrawn interval overlaps the delivered one (Appendix D). For the Condorcet requirement we report existence separately. On HELM a Condorcet winner exists on the observed matrix (it is the mean leader) and in 86% of 200 noise draws; on OLLB it does not exist on the observed matrix nor in any of 50 draws, so there the Condorcet cost is the cost to beat every rival on a majority of tasks, not to unseat an existing winner (boundary cases in Table 34, Appendix E).

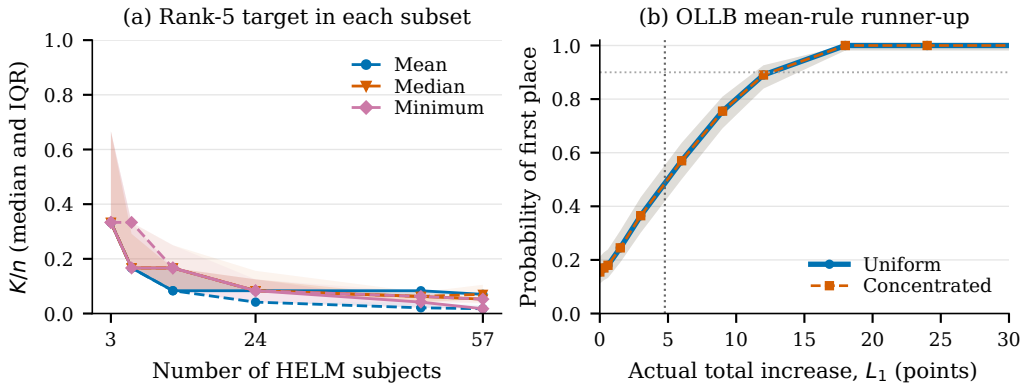


Figure 4: (a) Rule-specific rank-5 targets reselected in each HELM subset (50 subsets per size); solid/dashed curves use observed/shuffled scores, with subset IQR bands. (b) Fixed interventions on the OLLB mean-rule runner-up, ceiling cap; bands are Wilson 95% intervals over 200 noise draws. Dotted lines mark the nominal minimum L_1 and 90% success.

4.5 Near the top, evaluation noise matters

Near the top the investment is small: under the mean rule, the OLLB targets ranked 2 to 5 need between 4.8 and 23.9 points in total (HELM: 4.0 to 154.0), and the scores carry sampling noise from finite test sets. We therefore ask not for the re-optimised cost under noise, which is a cost distribution, but how likely a *fixed* investment is to work. We replay a uniform raise chosen on the observed leaderboard against $R = 200$ noise draws of all scores, modelled as Gaussian with the binomial standard error of each cell, clipped to $[0, 100]$ (Figure 4b for one target; Figures 8 and 9 for six ranks between 1 and 20). The statistics below are medians over 36 targets per dataset (ranks 2 to 11 and 26 log-spaced ranks up to the last), with intervals over targets (Appendix D). Raising every task by the nominal minimum L_∞ puts the target first in a median of 0.14 to 0.43 of the draws on OLLB (score rules) and 0.14 to 0.80 on HELM, except that this probability is knife-edge, since the raise is designed to pass the best rival by 10^{-3} points. A more stable quantity is the further uniform raise needed to reach 90%: a median of 1.3 to 2.5 points on OLLB and 0.3 to 2.0 on HELM for every rule but the minimum, and 5.6 (OLLB) and 9.0 (HELM) for the minimum. Evaluation noise at the observed scale therefore adds a few points per task to the nominal minimum, and the nominal minimum is not a safe investment. A different noise seed or three redrawn target samples change the median probability by at most 0.02 (OLLB) and 0.08 (HELM) and the extra raise by at most 0.4 points for the rules other than minimum and Borda. The binomial standard error covers only item-sampling noise, and no repeated evaluations exist to estimate more, so 1.5 and 2 times the standard error are sensitivity multipliers, not estimates: the mean rule's extra raise for 90% grows from 1.3 to 2.0 and 3.0 points on HELM and from 1.3 to 2.0 and 2.8 on OLLB, roughly in proportion; the minimum rule's reaches 12.0 on OLLB and exceeds the 16-point grid for 17 of 36 HELM targets, and only close pairs of other rules swap. Centring the noise on a shrunk truth (regression to the mean, a model with assumptions in Appendix D) *raises* the probability (0.26–0.82 on OLLB, 0.25–0.97 on HELM), so the observed-centred replay is the conservative reading. This supports C5 in its probabilistic form; cost below noise holds only for targets very near the top. The sizes of all noise runs (draws, targets, seeds) are listed in Appendix D.

4.6 Number of tasks and cross-task correlation

On HELM, in 50 random subsets of $n \in \{3, \ldots, 57\}$ subjects, the mean's L_1 for a rank-5 target rises from 6.1 at $n = 3$ to 154.0 at $n = 57$, roughly in proportion to n, but only to 49.4 when each task column is permuted over models, which removes cross-task correlation; the median's L_1 falls with n (4.3 at $n = 12$, 1.5 at $n = 57$) and K/n falls (Figure 4a; details in Appendix D). This compares curves with and without correlation, not fitted exponents, and subsets of MMLU subjects are not independent leaderboards.

4.7 ROBUSTNESS

The rule order of the median L_1 at ceil (minimum $<$ median $<$ harmonic $<$ geometric $<$ mean on OLLB) is the same in all 5 data slices we ran (all models; merges removed; top 500 models; raw instead of normalised scores; duplicates resolved by keeping the worst variant), although the magnitudes change (for example the minimum's median L_1 is 43 on all models and 75 on raw scores). A margin ε up to 0.1 changes any median L_1 at ceil by at most 2.0% (and the default $\varepsilon = 10^{-3}$ by far less); the offset of the geometric and harmonic means (0.1 to 10 against 1) by at most 8.2%. Random Dirichlet weights, bootstrap over tasks and noise draws for the targets studied are in Appendix C.

5 DISCUSSION AND LIMITATIONS

What the evidence says. Under our model the answer to "what investment puts a target first, and how does it vary with the rule" is conditional on three things. *The cost measure*: in total points the mean is among the dearest rules, and the minimum (on both leaderboards) and the median (on HELM) are several times cheaper, but not in the largest single gain, and in tasks touched the order is different and not stable (Sections 4.1, 4.3). *The cap*: whether first place is reachable at all under bounded per-task gains depends on the rule, and on OLLB for most targets the answer is no at 10 points under every rule (Section 4.2). *The target's position and the noise*: targets near the top need few points, but the nominal minimum succeeds in a median of only 0.14 to 0.43 (OLLB) and 0.14 to 0.80 (HELM) of noise draws (Section 4.5). Of the six pre-stated claims, C1 holds numerically, C2 is refuted as a general order, C3 is supported, C4 holds at the ceiling and is qualified by feasibility, C5 holds in its probabilistic form, and C6 compares curves with and without correlation.

Where the conclusions fail or do not apply. (i) *Cost is in leaderboard points*, not in data, compute or money; the caps stand in for what is plausible, and the empirical caps (emp95, emp99) are general fine-tuning gains on six tasks, not the effect of task-specific data. (ii) *Rivals are fixed* and tasks do not interact: no spillover of a gain to other tasks (which would lower the cost) and no response from other developers (which would raise it; the equilibrium question is studied by Chen et al. (2026)). (iii) *Two leaderboards*, both narrow: six heterogeneously normalised tasks on OLLB and one benchmark split into 57 subjects on HELM; the n-sweep varies n only within MMLU. (iv) *Rank-rule coverage*: the Condorcet requirement stands for Condorcet-consistent rules, not for Kemeny (not run); on OLLB rank rules run on a rank-stratified sample whose quantiles are not those of all targets. (v) *Normalisation*: OLLB scores are the leaderboard's normalised points; a raw-score slice keeps the order of rules but not the magnitudes. (vi) *Ties and discreteness*: ε is small and its effect bounded (Section 4.7). (vii) *Dependence*: intervals on target medians describe the set of targets studied, not independent samples of models and not evaluation uncertainty; bootstrapping OLLB over 3849 base-model families widens them by a median factor of 1.7 and changes no conclusion (Section 4.2), while HELM has no family labels. (viii) *Noise model*: Gaussian with item-sampling errors only; the overdispersion factors are sensitivity multipliers, not estimates, and the regression-to-the-mean reading rests on stated assumptions.

Unfinished or deviating work. The OLLB sample is a deterministic grid, so no seed applied to it; the first noise and bootstrap runs used the script default seed 0, not the study seed, and re-runs with the study seed and redrawn samples agree within the stated intervals. HELM rank-rule noise runs use 100 instead of 200 draws (exact optimisation cost), and the total compute exceeded the pre-run estimate of 4 CPU hours (Appendices A, D).

Implication. On these two leaderboards, and under our assumptions, a designer who worries about targeted investment should not infer robustness from the rule's name: whether the median or the minimum is harder to game than the mean depends on whether the developer pays in points or in tasks, and on how far a single score can plausibly move. Reporting a rule with an assumed largest gain per task, and the share of models that could reach first place within it, is a cheap check; the assumed gain itself is not something this study supplies.

Reproducibility statement

All code, results and run commands are in the accompanying repository (anonymised for review): `analysis/` holds the solvers, the entry point `run_all.sh`, the processed data tables and every result file with a manifest of checksums; `figures/` and `tables/` hold the scripts that produce each figure and table; `paper/scripts/make_numbers.py` regenerates every number quoted in the text. Data versions, checksums, seeds, run times and the licence judgement are in Appendix A and the repository `README.md`.

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A REPRODUCTION NOTES

Data and checksums. HELM MMLU release v1.13.0 (`mmlu/benchmark_output/releases/v1.13.0/`, `summary.json` date 2025-01-10), fetched by `analysis/src/fetch_helm.py`; SHA-256 of the file manifest `830bdb10a5518c50689d60c86623cc3d79ff5317d7e6b96408ea7dc2058de12b`. Open LLM Leaderboard `open-llm-leaderboard/contents`, commit `9c09a7cae43334062a82cb164f2ef255013dafa2` (2025-03-20), parquet file SHA-256 `f13a7061e63133bc8745d89f8c7a56148584940ede22d3b27e21cad1e078dca5`. The derived tables in `analysis/data/processed/` are rebuilt byte-identically from the downloads (SHA-256 in `manifest.json`); item counts for the noise model come from a single per-model results file of the `results` dataset (commit `aa81ecc38fdc5708254b833923368970efdf5ef5`, file SHA-256 in `ollb_noise.json`); no other file of that dataset and nothing from HELM `runs/` is used. No other data are used.

Licences. Neither source declares a licence for the scores: the `contents` dataset card has no licence field (it carries each model's own hub licence as a column), `results` has no card, and we found no separate statement for the HELM published results (the HELM code is Apache-2.0). Our judgement is that benchmark scores, as facts about public models, may be used for non-commercial academic analysis with attribution; we use no model weights, texts or prompts. The derived tables in the repository allow checking, but the raw sources remain the authority and anyone redistributing them should check the sources' terms.

Environment, seeds, time. Python 3.11.2, numpy 2.4.6, scipy 1.17.1 (HiGHS), pandas 3.0.6, pyarrow 26.0.0, pytest 9.2.0; 2 CPUs, 4 GB. The study seed is 20261108 (bootstrap over targets, weights, replay, Condorcet existence, n-sweep). The first evaluation-noise and task-bootstrap runs and the power-mean/offset/ε runs used the script default seed 0; the OLLB rank-stratified sample is a deterministic grid of ranks and does not depend on a seed; the replay on 36 targets, the seed checks and the cluster bootstrap use the study seed or the seeds named in their tables. The full run takes about 8 CPU-hours in total, the longest single job about 5.6 hours (OLLB score rules over seven caps), with jobs independent and each below 700 MB; this exceeds the 4 CPU-hour figure in the task, which was an unmeasured estimate. A fast mode (`bash analysis/run_all.sh fast`) rebuilds the data, runs the tests, the theory check and the HELM headline grid in about ten minutes.

Commands and script-to-output map. `bash analysis/run_all.sh full` produces every file under `analysis/results/`, each listed in `MANIFEST.csv` with its producing script, grid check and SHA-256 (46 of 46 files complete, 0 unresolved solver statuses). Numbers in the text: `python3 paper/scripts/make_numbers.py` writes `paper/numbers.tex` from the claim tables. Figures and tables: Table 3.

Table 3: Script-to-output map. Figures and tables made by `presentation/render.py` (display summaries only) are listed with the result files in `analysis/results/` they are made from, each followed by the analysis script that wrote it (`MANIFEST.csv`); where an asset has more than three result files the list is cut (`...`) and the exact list is in `presentation/manifest.json`. The two tables written by the report are listed last: their numbers come from the claim tables via `paper/scripts/make_numbers.py`.

Where	Asset	Result files (analysis script)
Fig. 7	`cap_map_helm`	`cap_rule_summary.csv` (claims.py)
Fig. 5	`cap_map_ollb`	`cap_rule_summary.csv` (claims.py)
Fig. 6	`cap_map_ollb_sample`	`cap_rule_summary.csv` (claims.py)
Fig. 3	`cap_map`	`cap_rule_summary.csv` (claims.py)
Fig. 14	`cluster_uncertainty`	`cluster_boot.csv` (cluster_boot.py)
Fig. 12	`gstar_matched`	`gstar_curves.csv` (gstar.py), `gstar_summary.csv` (gstar.py)
Fig. 2	`gstar_curves`	`gstar_curves.csv` (gstar.py), `gstar_summary.csv` (gstar.py)
Fig. 4	`scaling_and_noise`	`robust_nsweep.csv` (robust_summary.py), `ollb_ceil.csv` (run_replay.py)

Where	Asset	Result files (analysis script)
Fig. 15	normalised_rule_costs	abstract_ratio_check.csv (normalised_costs.py), rank_matched.csv (normalised_costs.py)
Fig. 1	cost_by_rank	ollb_scores_caps.csv (run_core.py ollb), ollb_trimmed1.csv (run_core.py ollb trimmed1), ollb_ordinal_sample.csv (run_core.py ollb ... sample)
Fig. 13	replay_extra_raise	replay2_summary.csv (replay2_summary.py)
Fig. 9	replay_helm	helm_ceil.csv (run_replay.py), helm_gain10.csv (run_replay.py)
Fig. 8	replay_ollb	ollb_ceil.csv (run_replay.py), ollb_gain10.csv (run_replay.py)
Fig. 11	task_scaling_gain10	tasksub_helm.csv (run_robust.py tasksub [--shuffle])
Fig. 10	task_scaling_ceil	tasksub_helm.csv (run_robust.py tasksub [--shuffle])
Tab. 25	cluster_family_sensitivity	cluster_boot.csv (cluster_boot.py)
Tab. 24	cluster_headline	cluster_boot.csv (cluster_boot.py)
Tab. 28	cluster_sizes	cluster_sizes_top30.csv (cluster_boot.py), cluster_sizes.csv (cluster_boot.py)
Tab. 26	cluster_strata_costs	cluster_boot.csv (cluster_boot.py)
Tab. 27	cluster_strata_feasibility	cluster_boot.csv (cluster_boot.py)
Tab. 7	costs_full_helm	cap_rule_summary.csv (claims.py)
Tab. 5	costs_full_ollb	cap_rule_summary.csv (claims.py)
Tab. 6	costs_full_ollb_sample	cap_rule_summary.csv (claims.py)
Tab. 34	counterexamples	condorcet_existence.csv (condorcet_exist.py), cap_rule_summary.csv (claims.py), c2_order_statistics.csv (claims.py)
Tab. 16	robust_eps	robust_eps.csv (robust_summary.py)
Tab. 20	gstar_gain_anchor	gstar_anchor_finetune_gains.csv (gstar.py)
Tab. 19	gstar_thresholds	gstar_summary.csv (gstar.py)
Tab. 13	mean_leader_helm	ollb_closed.csv (run_core.py ollb), ollb_scores_caps.csv (run_core.py ollb), ollb_trimmed1.csv (run_core.py ollb trimmed1), ...
Tab. 12	mean_leader_ollb	ollb_closed.csv (run_core.py ollb), ollb_scores_caps.csv (run_core.py ollb), ollb_trimmed1.csv (run_core.py ollb trimmed1), ...
Tab. 31	minimum_zero_diagnostic	min_zero_diagnostic.csv (nozero_diag.py)
Tab. 29	normalised_cost_ratios	normalised_summary.csv (normalised_costs.py)
Tab. 15	robust_offsets	robust_offsets.csv (robust_summary.py)
Tab. 9	order_statistics	c2_order_statistics.csv (claims.py)
Tab. 32	ordinal_seed_ceil	seed_stability.csv (seed_stability.py)
Tab. 33	ordinal_seed_gain10	seed_stability.csv (seed_stability.py)
Tab. 8	ordinal_comparison	c4_ordinal_vs_mean.csv (claims.py)
Tab. 11	power_means_helm	helm_powmean.csv (run_core.py helm)
Tab. 10	power_means_ollb	ollb_powmean.csv (run_core.py ollb)
Tab. 30	rank_matched_ratios	rank_matched.csv (normalised_costs.py)
Tab. 23	reoptimised_seed_sensitivity	noise_seed_check.csv (noise_seed_check.py)
Tab. 17	robust_reoptimized_costs	robust_noise_taskboot_per_target.csv (robust_summary.py)
Tab. 21	replay_extended	replay2_summary.csv (replay2_summary.py)
Tab. 22	replay_seed_sensitivity	replay2_summary.csv (replay2_summary.py)
Tab. 14	robust_slices	robust_slices.csv (slice_summary.py)
Tab. 2	main_summary	cap_rule_summary.csv (claims.py)
Tab. 4	theory_validation	theory_validation_summary.json (theory_validate.py), theory_validation.csv (theory_validate.py)
Tab. 18	robust_weights	robust_weights.csv (robust_summary.py)
Tab. 1	claims	cap_rule_summary.csv, c2_order_statistics.csv, c4_ordinal_vs_mean.csv, replay2_summary.csv; typeset in paper/sections/experiments.tex; values via paper/scripts/make_numbers.py

Where	Asset	Result files (analysis script)
Tab. 3	`script map`	`presentation/manifest.json, MANIFEST.csv; paper/scripts/make_map.py`

Literature and forbidden material. The task withholds one published paper that studied this question, together with any page that cites, discusses or reproduces it. We did not search for it, and we did not guess its title, authors or results. The writing position's literature searches (titles and abstracts through OpenAlex and the arXiv listing of the items in the starter list) were restricted to social choice, bribery and margin of victory, ranking robustness, contamination and benchmark aggregation. We met no content that we recognised as the withheld paper or as a discussion of it, and read none. The context position reports the same: it never searched for the withheld paper, scoped its searches to the starter list and adjacent topics (social choice, bribery and margin of victory, ranking robustness, contamination, the benchmark papers), met nothing it could identify as the withheld paper, and opened no full text. The analysis position (data downloads only; no web or literature search), the plan position (no searches) and the venue position (conference format pages and one arXiv query for a style file, which returned 15 unrelated titles and abstracts; the sources of 6 of them were refused and no content was read) each report that they did not search for, meet or read the withheld paper. The presentation position likewise reports that it did not search for, meet or read it, and worked from the objective, permitted context, committed results and venue material without external literature searches. All references were read at the level of title and abstract only, and the paper's statements about them stay at that level. The log is `context/FORBIDDEN_LOG.md`.

B CLOSED FORMS AND THEIR CHECK

For the mean, with $R = nb - \sum_t x_t$ and $h_t = c_t - x_t$: $L_1 = R$ when $\sum_t h_t \geq R$ (and does not depend on how the points are placed); K is the least k for which the k largest headrooms sum to at least R; L_∞ is the level d with $\sum_t \min(d, h_t) = R$. For the minimum: $K = |\{t : x_t < b\}|$, $L_1 = \sum_t (b - x_t)^+$, $L_\infty = b - \min_t x_t$, infeasible iff some $x_t < b$ has $c_t < b$. For the median at the ceiling cap, raising the k lowest tasks to 100 is optimal, and for odd n with $c = |\{t : x_t \geq b\}|$, $K = \max(0, (n+1)/2 - c)$. K equals $\lfloor n/2 \rfloor + 1$ for only a small share of targets, so the blanket statements "the median needs $\lceil n/2 \rceil + 1$ tasks" and "the minimum needs all n tasks" are false. Table 4 compares these with the solvers on every target at the ceiling cap. The agreement is numerical: 0 mismatches in 16 cells. The closed forms are validated only at ceil; with general caps one replaces 100 by c_t, and the solver is what we ran. For even n (OLLB) only K of the median has a closed form.

Table 4: Ceiling-cap closed-form consistency checks on all mean nonleaders. Maximum absolute discrepancy versus the solver; zero mismatches (exact K; L_1 tolerance 10^{-6}, L_∞ tolerance 2×10^{-6}). Even-n median L_1/L_∞ has no checked closed form here.

Dataset / rule	Metric	N	Max. error	Mismatches
OLLB / Mean	K	4495	0	0
OLLB / Mean	L_1	4495	5.7×10^{-14}	0
OLLB / Mean	L_∞	4495	9.3×10^{-8}	0
OLLB / Minimum	K	4495	0	0
OLLB / Minimum	L_1	4495	1.4×10^{-14}	0
OLLB / Minimum	L_∞	4495	9.1×10^{-8}	0
OLLB / Median	K	4495	0	0
HELM / Mean	K	78	0	0
HELM / Mean	L_1	78	2.3×10^{-13}	0
HELM / Mean	L_∞	78	9.2×10^{-8}	0
HELM / Minimum	K	78	0	0
HELM / Minimum	L_1	78	2.3×10^{-13}	0
HELM / Minimum	L_∞	78	9.2×10^{-8}	0
HELM / Median	K	78	0	0
HELM / Median	L_1	78	1.1×10^{-13}	0
HELM / Median	L_∞	78	9.4×10^{-8}	0

C ROBUSTNESS DETAILS

Overview. Every figure and table below is produced by `presentation/render.py` from the files listed in Table 3. Full cap maps, the cost tables for every cap and rule, and the replay and task-count figures follow; the target sets are the OLLB census (4495 targets, score rules), the OLLB rank-stratified sample (136 targets, all rules) and HELM (78 targets).

FIGURES

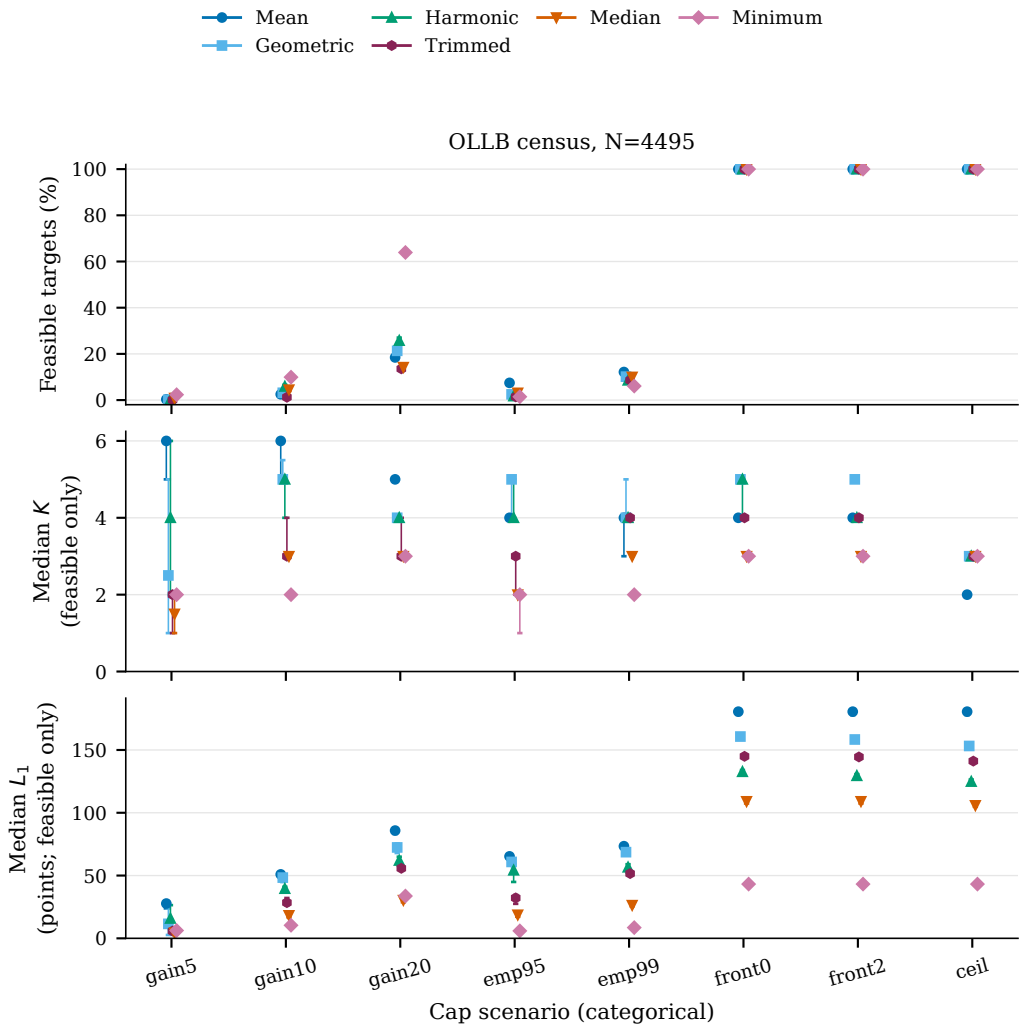


Figure 5: Full cap map for OLLB census ($N = 4495$). Feasibility and separately minimized costs; markers are separate cap scenarios, with no connecting lines. Interval conventions match the main cap map.

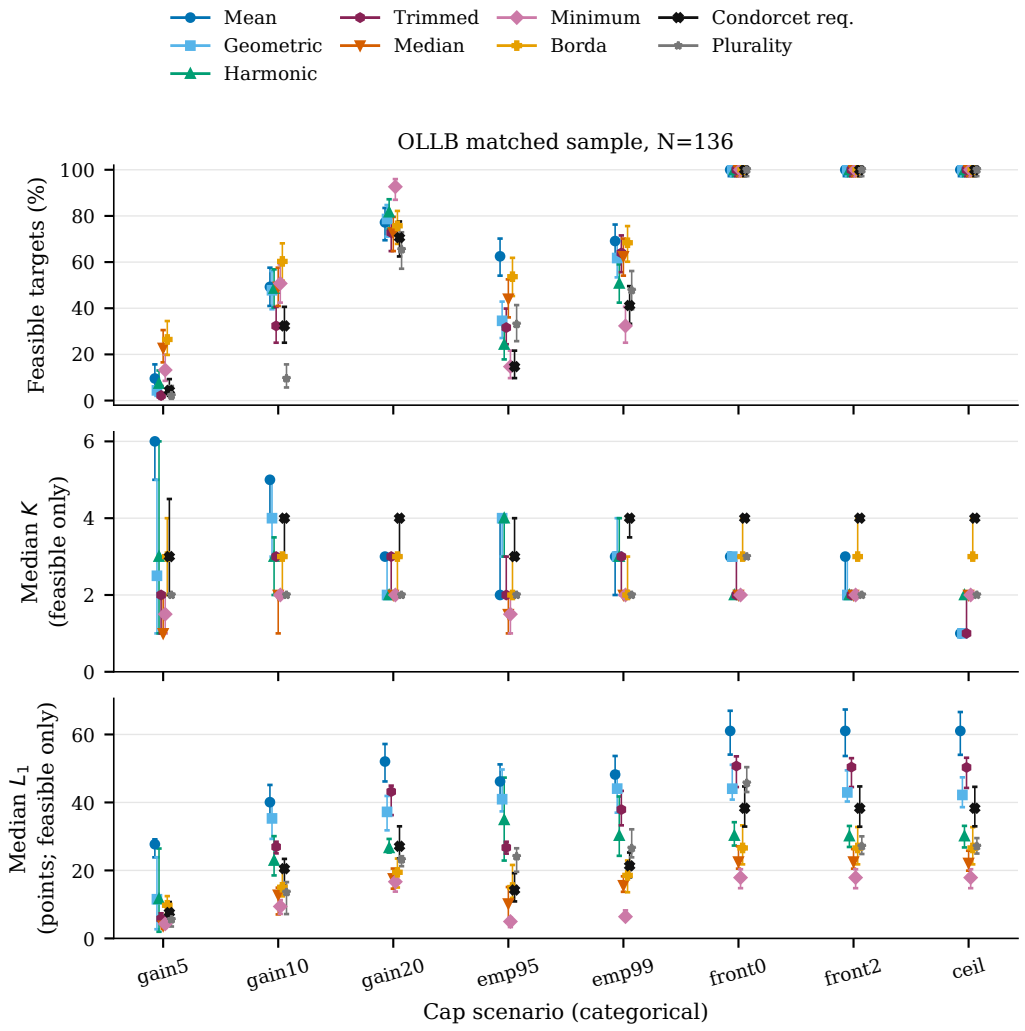


Figure 6: Full cap map for OLLB matched sample ($N = 136$). Feasibility and separately minimized costs; markers are separate cap scenarios, with no connecting lines. Interval conventions match the main cap map.

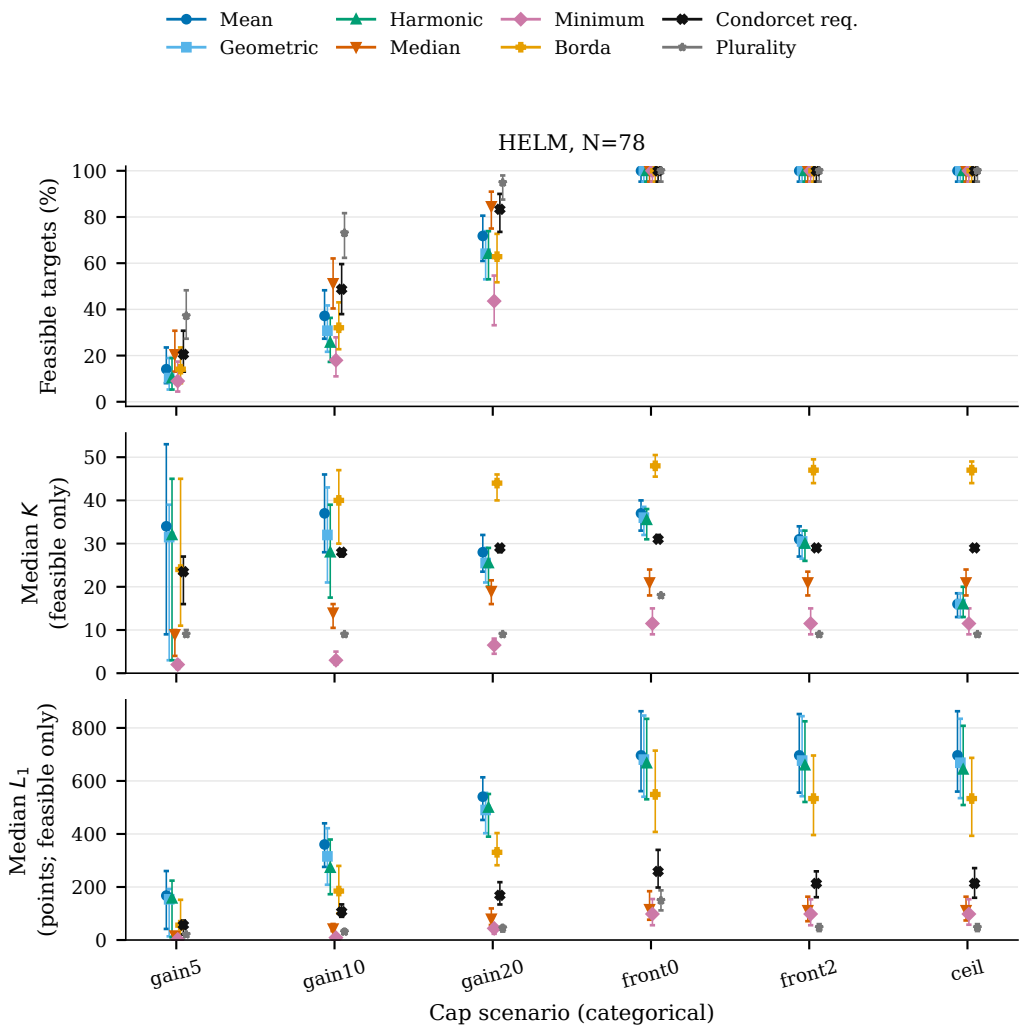


Figure 7: Full cap map for HELM ($N = 78$). Feasibility and separately minimized costs; markers are separate cap scenarios, with no connecting lines. Interval conventions match the main cap map.

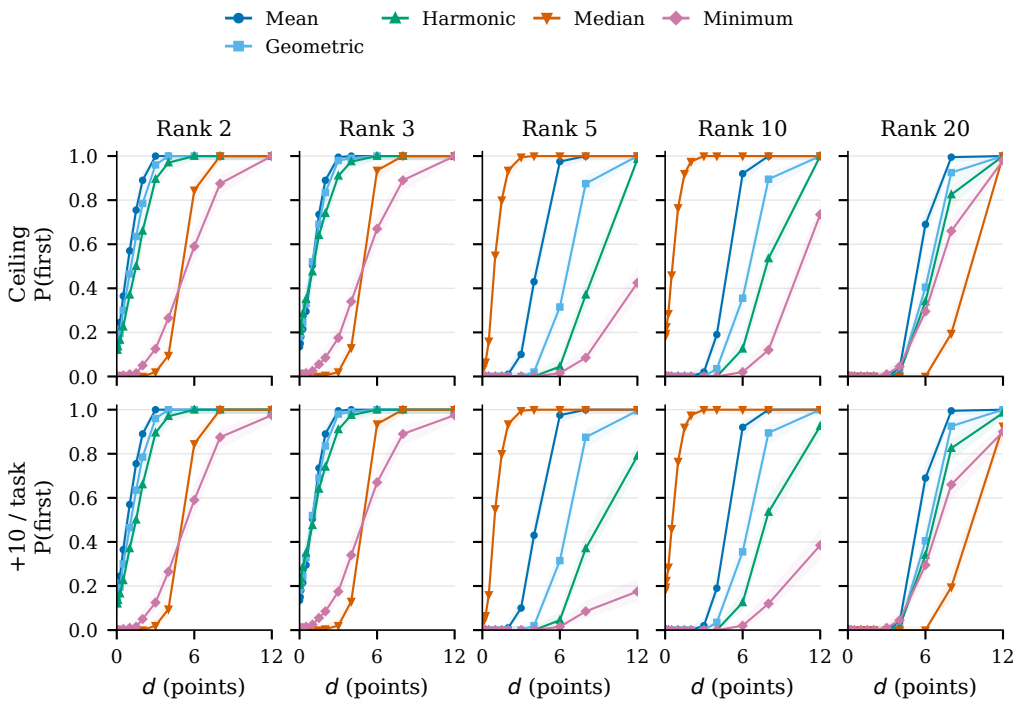


Figure 8: Uniform fixed-strategy replay on OLLB: pointwise Wilson 95% intervals over 200 noise draws; target ranks are fixed by the observed mean rule.

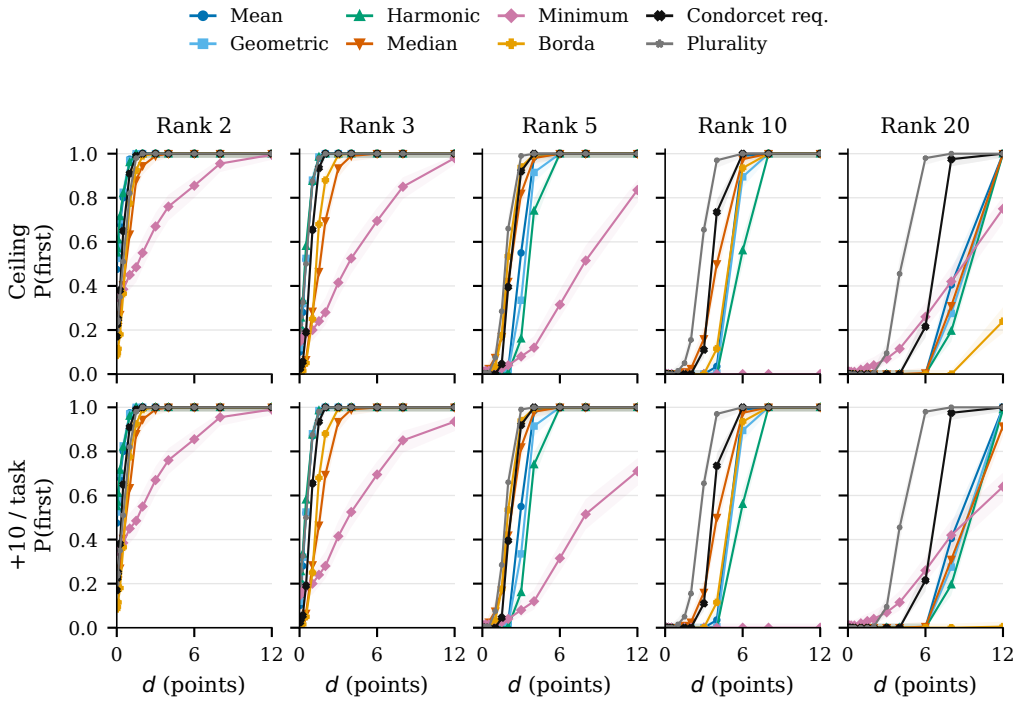


Figure 9: Uniform fixed-strategy replay on HELM: pointwise Wilson 95% intervals over 200 noise draws; target ranks are fixed by the observed mean rule.

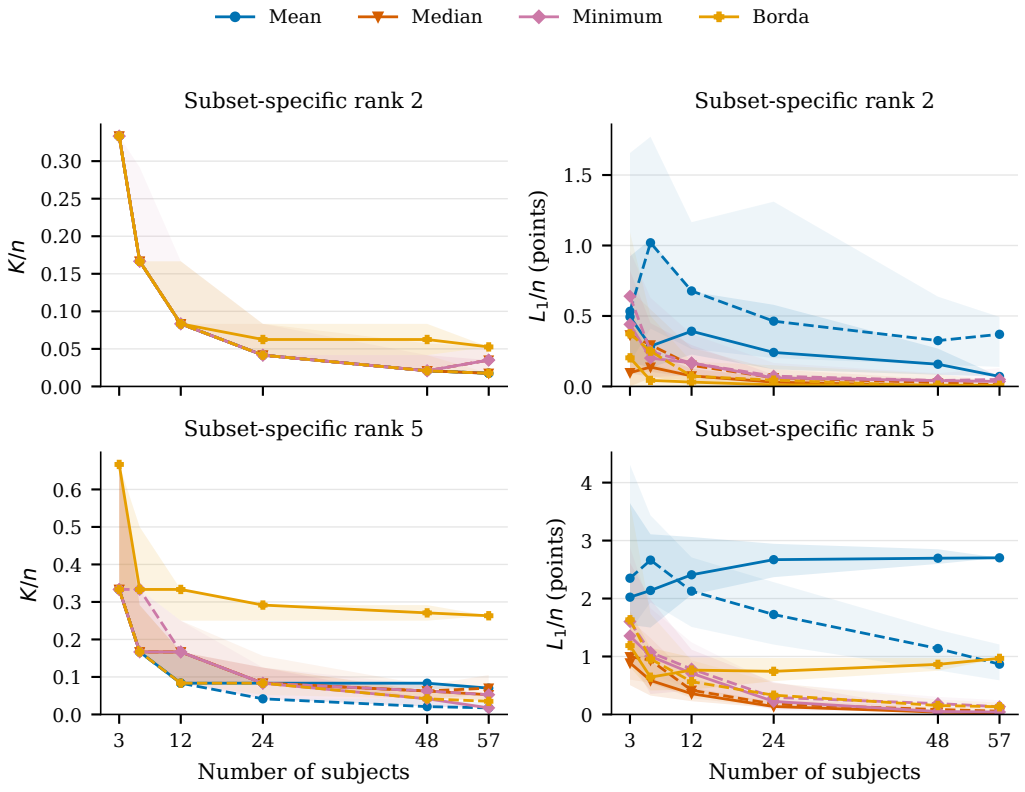


Figure 10: HELM task-subset scaling under ceil: medians and IQRs across 50 subset draws per size; costs condition on feasible draws.

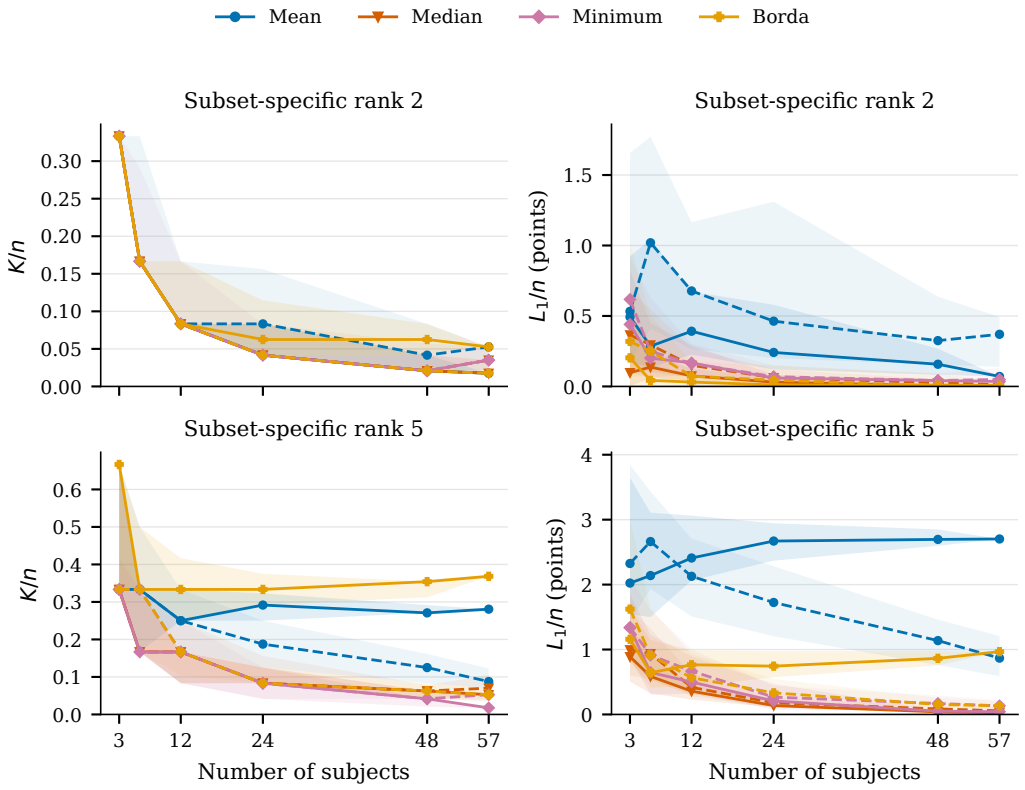


Figure 11: HELM task-subset scaling under gain10: medians and IQRs across 50 subset draws per size; costs condition on feasible draws.

TABLES

Table 5: Full OLLB census cap/rule costs: median [95% target-bootstrap interval], conditional on feasibility; finite/total counts are shown.

Rule	Feasible	K	L_1	L_∞
gain5				
Mean	13/4495	6.0 [5.0, 6.0]	27.7 [23.9, 29.1]	4.6 [4.0, 4.9]
Geometric	6/4495	2.5 [1.0, 5.0]	11.5 [2.7, 23.9]	2.5 [1.0, 4.4]
Harmonic	13/4495	4.0 [2.0, 6.0]	15.8 [7.5, 26.4]	3.9 [2.5, 4.8]
Trimmed	3/4495	2.0 [1.0, 2.0]	5.8 [4.6, 7.4]	1.4 [1.2, 1.8]
Median	40/4495	1.5 [1.0, 2.0]	4.4 [3.2, 5.7]	2.2 [1.6, 2.8]
Minimum	105/4495	2.0 [2.0, 2.0]	6.2 [5.8, 6.8]	4.1 [4.0, 4.4]
gain10				
Mean	113/4495	6.0 [5.0, 6.0]	50.9 [47.5, 52.6]	8.5 [7.9, 8.8]
Geometric	142/4495	5.0 [5.0, 5.5]	48.4 [44.8, 49.9]	8.8 [8.5, 9.1]
Harmonic	264/4495	5.0 [4.0, 5.0]	39.5 [37.9, 41.0]	8.5 [8.4, 8.6]
Trimmed	55/4495	3.0 [3.0, 4.0]	28.4 [26.4, 32.1]	7.1 [6.6, 7.9]
Median	203/4495	3.0 [3.0, 3.0]	18.3 [16.8, 20.5]	8.2 [7.8, 8.4]
Minimum	446/4495	2.0 [2.0, 2.0]	10.4 [9.8, 11.0]	6.6 [6.3, 6.7]
gain20				
Mean	830/4495	5.0 [5.0, 5.0]	85.8 [83.2, 88.2]	14.3 [13.9, 14.7]
Geometric	959/4495	4.0 [4.0, 4.0]	72.4 [67.9, 75.5]	14.1 [13.6, 14.8]
Harmonic	1157/4495	4.0 [4.0, 4.0]	62.0 [57.7, 65.0]	15.0 [14.5, 15.3]
Trimmed	609/4495	3.0 [3.0, 4.0]	55.8 [54.5, 56.9]	13.7 [13.5, 14.0]
Median	641/4495	3.0 [3.0, 3.0]	30.4 [28.8, 31.9]	12.3 [11.9, 12.8]
Minimum	2873/4495	3.0 [3.0, 3.0]	33.7 [33.1, 34.3]	15.7 [15.5, 15.8]
emp95				
Mean	335/4495	4.0 [4.0, 4.0]	65.3 [64.1, 66.8]	22.5 [21.6, 24.0]
Geometric	111/4495	5.0 [4.0, 5.0]	60.8 [51.3, 65.6]	20.0 [16.4, 22.9]
Harmonic	77/4495	4.0 [4.0, 5.0]	54.2 [44.9, 59.2]	16.4 [13.2, 20.6]
Trimmed	64/4495	3.0 [2.0, 3.0]	32.2 [27.5, 34.5]	13.3 [10.3, 16.5]
Median	141/4495	2.0 [2.0, 2.0]	18.8 [17.2, 20.9]	11.6 [9.6, 13.2]
Minimum	65/4495	2.0 [1.0, 2.0]	5.9 [4.2, 7.0]	3.8 [3.7, 4.8]
emp99				
Mean	546/4495	4.0 [3.0, 4.0]	73.4 [71.0, 74.9]	17.9 [16.8, 18.7]
Geometric	456/4495	4.0 [4.0, 5.0]	68.6 [66.9, 70.4]	16.6 [15.8, 17.6]
Harmonic	384/4495	4.0 [4.0, 4.0]	56.6 [52.4, 59.0]	11.7 [11.2, 12.7]
Trimmed	387/4495	4.0 [4.0, 4.0]	51.6 [51.0, 52.8]	19.3 [18.5, 20.0]
Median	451/4495	3.0 [3.0, 3.0]	26.4 [25.5, 27.5]	14.1 [13.5, 14.7]
Minimum	273/4495	2.0 [2.0, 2.0]	8.5 [7.9, 9.0]	5.4 [5.1, 5.6]
front0				
Mean	4495/4495	4.0 [4.0, 4.0]	180.5 [178.0, 182.3]	32.8 [32.5, 33.2]
Geometric	4495/4495	5.0 [5.0, 5.0]	160.7 [158.1, 162.6]	29.9 [29.5, 30.1]
Harmonic	4495/4495	5.0 [4.0, 5.0]	132.6 [130.7, 134.0]	26.0 [25.7, 26.2]
Trimmed	4495/4495	4.0 [4.0, 4.0]	144.9 [143.3, 146.3]	35.6 [35.2, 36.0]
Median	4495/4495	3.0 [3.0, 3.0]	109.1 [107.4, 110.5]	37.4 [36.9, 37.9]
Minimum	4495/4495	3.0 [3.0, 3.0]	43.2 [42.5, 44.1]	17.9 [17.7, 18.1]
front2				
Mean	4495/4495	4.0 [4.0, 4.0]	180.5 [177.8, 182.3]	31.9 [31.5, 32.2]
Geometric	4495/4495	5.0 [5.0, 5.0]	158.4 [156.0, 160.3]	28.8 [28.6, 29.1]
Harmonic	4495/4495	4.0 [4.0, 4.0]	129.4 [127.5, 130.8]	25.2 [24.9, 25.5]
Trimmed	4495/4495	4.0 [4.0, 4.0]	144.5 [142.9, 145.7]	35.1 [34.7, 35.5]
Median	4495/4495	3.0 [3.0, 3.0]	109.1 [107.4, 110.3]	37.3 [36.9, 37.8]
Minimum	4495/4495	3.0 [3.0, 3.0]	43.2 [42.4, 44.1]	17.9 [17.7, 18.1]
ceil				
Mean	4495/4495	2.0 [2.0, 2.0]	180.5 [177.8, 182.3]	30.1 [29.7, 30.4]
Geometric	4495/4495	3.0 [3.0, 3.0]	153.2 [150.8, 154.8]	27.8 [27.5, 28.0]
Harmonic	4495/4495	3.0 [3.0, 3.0]	124.8 [123.7, 126.5]	24.9 [24.6, 25.1]
Trimmed	4495/4495	3.0 [3.0, 3.0]	141.1 [139.8, 142.5]	33.7 [33.3, 34.0]
Median	4495/4495	3.0 [3.0, 3.0]	105.9 [104.8, 107.4]	36.2 [35.7, 36.5]
Minimum	4495/4495	3.0 [3.0, 3.0]	43.2 [42.4, 44.1]	17.9 [17.8, 18.1]

Table 6: Full OLLB matched sample cap/rule costs: median [95% target-bootstrap interval], conditional on feasibility; finite/total counts are shown.

Rule	Feasible	K	L_1	L_∞
gain5				
Mean	13/136	6.0 [5.0, 6.0]	27.7 [23.9, 29.1]	4.6 [4.0, 4.9]
Geometric	6/136	2.5 [1.0, 5.0]	11.5 [2.7, 23.9]	2.5 [0.9, 4.4]
Harmonic	10/136	3.0 [1.0, 6.0]	11.6 [2.0, 26.4]	3.2 [1.0, 4.8]
Trimmed	3/136	2.0 [1.0, 2.0]	5.8 [4.6, 7.4]	1.4 [1.2, 1.8]
Median	31/136	1.0 [1.0, 2.0]	3.7 [2.9, 4.8]	1.9 [1.4, 2.4]
Minimum	18/136	1.5 [1.0, 2.0]	4.3 [3.1, 6.1]	3.5 [2.9, 4.1]
Borda	36/136	3.0 [2.0, 4.0]	9.7 [7.2, 12.4]	3.3 [2.6, 3.9]
Condorcet req.	6/136	3.0 [2.0, 4.5]	7.6 [5.8, 10.7]	3.6 [2.5, 4.0]
Plurality	3/136	2.0 [2.0, 2.0]	5.5 [3.5, 7.5]	3.2 [2.2, 4.1]
gain10				
Mean	67/136	5.0 [4.0, 5.0]	40.1 [35.1, 45.2]	6.7 [5.9, 7.5]
Geometric	65/136	4.0 [3.0, 5.0]	35.3 [29.3, 40.4]	7.2 [6.8, 8.0]
Harmonic	66/136	3.0 [2.0, 3.5]	22.9 [18.5, 30.1]	7.2 [6.7, 8.0]
Trimmed	44/136	3.0 [3.0, 3.0]	26.9 [25.0, 28.4]	6.7 [6.3, 7.1]
Median	67/136	2.0 [1.0, 2.0]	12.8 [7.1, 14.9]	5.7 [3.5, 7.0]
Minimum	69/136	2.0 [2.0, 2.0]	9.4 [7.2, 11.3]	5.9 [5.5, 7.3]
Borda	82/136	3.0 [2.0, 3.0]	15.1 [12.4, 20.0]	5.4 [4.7, 5.8]
Condorcet req.	44/136	4.0 [3.0, 4.0]	20.5 [15.6, 23.4]	6.8 [6.0, 7.8]
Plurality	13/136	2.0 [2.0, 2.0]	13.6 [7.2, 16.6]	8.4 [6.0, 9.1]
gain20				
Mean	105/136	3.0 [3.0, 3.0]	52.0 [46.2, 57.2]	8.7 [7.8, 9.5]
Geometric	107/136	2.0 [2.0, 3.0]	37.2 [31.8, 41.9]	9.1 [8.2, 9.8]
Harmonic	111/136	2.0 [2.0, 2.0]	26.6 [25.3, 29.3]	9.6 [8.6, 10.0]
Trimmed	99/136	3.0 [2.0, 3.0]	43.2 [36.3, 44.9]	10.8 [9.1, 11.2]
Median	99/136	2.0 [2.0, 2.0]	17.7 [14.6, 20.5]	8.2 [6.7, 9.3]
Minimum	126/136	2.0 [2.0, 2.0]	16.7 [13.8, 18.8]	9.3 [8.3, 10.7]
Borda	103/136	3.0 [2.0, 3.0]	19.5 [15.0, 23.5]	6.0 [5.5, 7.2]
Condorcet req.	96/136	4.0 [3.0, 4.0]	27.1 [23.6, 33.0]	11.4 [9.0, 12.3]
Plurality	89/136	2.0 [2.0, 2.0]	23.2 [21.2, 24.2]	14.4 [12.9, 15.2]
emp95				
Mean	85/136	2.0 [2.0, 4.0]	46.2 [40.1, 51.2]	12.4 [9.3, 14.9]
Geometric	47/136	4.0 [3.0, 4.0]	41.0 [37.4, 49.7]	10.6 [9.1, 15.2]
Harmonic	33/136	4.0 [3.0, 4.0]	34.9 [22.9, 47.3]	10.7 [6.1, 15.5]
Trimmed	43/136	2.0 [2.0, 3.0]	26.7 [24.9, 28.4]	9.8 [8.3, 11.5]
Median	60/136	1.5 [1.0, 2.0]	10.3 [5.1, 15.1]	4.9 [2.5, 10.4]
Minimum	20/136	1.5 [1.0, 2.0]	5.0 [3.3, 6.1]	3.6 [3.1, 4.8]
Borda	73/136	2.0 [2.0, 3.0]	15.0 [11.9, 21.6]	5.5 [4.6, 6.7]
Condorcet req.	20/136	3.0 [3.0, 4.0]	14.2 [10.9, 19.2]	8.4 [5.4, 10.1]
Plurality	45/136	2.0 [2.0, 2.0]	24.1 [19.6, 26.5]	17.1 [12.9, 17.7]
emp99				
Mean	94/136	3.0 [2.0, 3.0]	48.2 [43.3, 53.7]	8.7 [7.5, 10.1]
Geometric	84/136	3.0 [3.0, 4.0]	44.1 [37.0, 49.1]	9.3 [7.9, 10.3]
Harmonic	69/136	3.0 [3.0, 4.0]	30.2 [24.3, 41.8]	8.1 [7.1, 10.0]
Trimmed	87/136	3.0 [2.0, 3.0]	37.9 [33.3, 43.4]	11.0 [9.3, 13.0]
Median	85/136	2.0 [2.0, 2.0]	15.6 [13.7, 18.3]	7.2 [6.4, 9.6]
Minimum	44/136	2.0 [2.0, 2.0]	6.4 [5.8, 8.2]	5.3 [4.6, 5.8]
Borda	93/136	2.0 [2.0, 3.0]	18.5 [13.6, 22.9]	5.7 [5.1, 6.8]
Condorcet req.	56/136	4.0 [3.5, 4.0]	21.3 [18.0, 25.3]	8.6 [7.4, 10.0]
Plurality	65/136	2.0 [2.0, 2.0]	26.5 [23.9, 32.1]	17.8 [17.2, 19.3]

Rule	Feasible	K	L_1	L_∞
front0				
Mean	136/136	3.0 [3.0, 3.0]	61.0 [54.1, 67.0]	10.3 [9.7, 11.4]
Geometric	136/136	3.0 [3.0, 3.0]	44.0 [40.8, 51.1]	10.3 [9.3, 11.3]
Harmonic	136/136	2.0 [2.0, 2.0]	30.2 [27.3, 34.2]	10.1 [9.6, 11.4]
Trimmed	136/136	2.0 [2.0, 3.0]	50.7 [44.5, 53.6]	12.7 [11.2, 13.3]
Median	136/136	2.0 [2.0, 2.0]	22.6 [20.5, 27.1]	10.8 [9.3, 12.0]
Minimum	136/136	2.0 [2.0, 2.0]	17.9 [14.8, 20.3]	9.9 [8.7, 11.1]
Borda	136/136	3.0 [3.0, 4.0]	26.6 [21.8, 33.3]	8.3 [6.7, 9.7]
Condorcet req.	136/136	4.0 [4.0, 4.0]	38.3 [32.9, 44.7]	13.7 [12.3, 14.3]
Plurality	136/136	3.0 [3.0, 3.0]	45.7 [43.1, 50.4]	19.8 [18.4, 21.0]
front2				
Mean	136/136	3.0 [2.0, 3.0]	61.0 [53.7, 67.3]	10.2 [9.3, 11.2]
Geometric	136/136	2.0 [2.0, 3.0]	42.9 [40.3, 49.4]	10.3 [9.3, 11.3]
Harmonic	136/136	2.0 [2.0, 2.0]	30.1 [26.9, 33.1]	10.1 [9.6, 11.3]
Trimmed	136/136	2.0 [2.0, 2.0]	50.4 [44.5, 53.0]	12.6 [11.2, 13.4]
Median	136/136	2.0 [2.0, 2.0]	22.6 [20.5, 26.3]	10.8 [9.3, 11.9]
Minimum	136/136	2.0 [2.0, 2.0]	17.9 [14.8, 20.3]	9.9 [8.8, 11.1]
Borda	136/136	3.0 [3.0, 4.0]	26.6 [21.8, 32.8]	8.3 [6.7, 9.7]
Condorcet req.	136/136	4.0 [4.0, 4.0]	38.3 [32.9, 44.7]	13.7 [12.3, 14.4]
Plurality	136/136	2.0 [2.0, 2.0]	27.1 [24.8, 30.0]	17.1 [15.3, 18.3]
ceil				
Mean	136/136	1.0 [1.0, 1.0]	61.0 [54.0, 66.6]	10.2 [9.0, 11.1]
Geometric	136/136	1.0 [1.0, 1.0]	42.2 [38.6, 47.4]	10.3 [9.4, 11.3]
Harmonic	136/136	2.0 [2.0, 2.0]	30.1 [26.8, 33.1]	10.1 [9.6, 11.2]
Trimmed	136/136	1.0 [1.0, 2.0]	50.3 [44.3, 53.1]	12.6 [11.1, 13.3]
Median	136/136	2.0 [2.0, 2.0]	22.2 [19.9, 25.8]	10.5 [9.2, 11.3]
Minimum	136/136	2.0 [2.0, 2.0]	17.9 [14.8, 20.3]	9.9 [8.7, 11.1]
Borda	136/136	3.0 [3.0, 4.0]	26.6 [21.8, 32.7]	8.3 [6.7, 9.7]
Condorcet req.	136/136	4.0 [4.0, 4.0]	38.3 [33.0, 44.6]	13.7 [12.3, 14.4]
Plurality	136/136	2.0 [2.0, 2.0]	27.1 [25.0, 29.5]	17.1 [15.3, 18.2]

Table 7: Full HELM cap/rule costs: median [95% target-bootstrap interval], conditional on feasibility; finite/total counts are shown.

Rule	Feasible	K	L_1	L_∞
gain5				
Mean	11/78	34.0 [9.0, 53.0]	167.6 [41.8, 260.2]	3.0 [0.7, 4.6]
Geometric	8/78	31.5 [3.0, 39.0]	153.6 [13.9, 192.6]	2.9 [0.3, 3.6]
Harmonic	8/78	32.0 [3.0, 45.0]	157.1 [10.0, 224.0]	3.2 [0.6, 4.2]
Median	16/78	9.0 [4.0, 10.0]	16.6 [3.3, 20.7]	2.9 [1.3, 3.8]
Minimum	7/78	2.0 [1.0, 3.0]	2.4 [2.0, 7.6]	2.4 [1.2, 4.2]
Borda	11/78	24.0 [11.0, 45.0]	56.0 [21.1, 151.7]	2.0 [1.2, 4.0]
Condorcet req.	16/78	23.5 [16.0, 27.0]	56.5 [20.7, 73.5]	3.1 [1.6, 4.0]
Plurality	29/78	9.0 [9.0, 10.0]	20.7 [15.2, 25.3]	3.0 [2.5, 3.5]
gain10				
Mean	29/78	37.0 [28.0, 46.0]	360.4 [275.7, 440.2]	6.5 [4.9, 8.2]
Geometric	24/78	32.0 [21.0, 43.0]	315.4 [208.5, 421.7]	6.0 [4.3, 8.1]
Harmonic	20/78	28.0 [17.5, 39.0]	272.8 [172.7, 379.0]	5.8 [4.2, 7.6]
Median	40/78	14.0 [10.5, 16.0]	43.0 [33.8, 60.5]	5.8 [4.8, 7.0]
Minimum	14/78	3.0 [2.0, 5.0]	9.0 [2.7, 18.5]	5.1 [2.7, 7.6]
Borda	25/78	40.0 [30.0, 47.0]	184.7 [100.5, 279.5]	5.5 [3.1, 7.4]
Condorcet req.	38/78	28.0 [27.0, 28.5]	107.0 [87.5, 134.3]	5.8 [4.0, 6.8]
Plurality	57/78	9.0 [9.0, 9.0]	31.0 [25.7, 40.4]	4.9 [3.8, 5.5]

Rule	Feasible	K	L_1	L_∞
gain20				
Mean	56/78	28.0 [23.5, 32.0]	540.6 [452.8, 613.7]	9.9 [8.5, 11.5]
Geometric	50/78	25.5 [21.0, 29.0]	490.5 [402.9, 555.1]	10.4 [8.2, 11.1]
Harmonic	50/78	25.5 [20.0, 29.0]	499.7 [389.9, 550.8]	11.2 [9.1, 12.0]
Median	66/78	19.0 [16.0, 21.5]	80.1 [56.7, 119.1]	8.3 [6.8, 10.3]
Minimum	34/78	6.5 [4.5, 8.0]	44.0 [23.4, 49.7]	11.7 [8.2, 15.6]
Borda	49/78	44.0 [40.0, 46.0]	330.8 [282.1, 403.4]	10.0 [7.4, 11.4]
Condorcet req.	65/78	29.0 [28.0, 29.0]	168.5 [134.1, 218.2]	8.3 [6.4, 10.3]
Plurality	74/78	9.0 [9.0, 9.0]	45.5 [31.7, 53.8]	6.1 [4.9, 7.2]
front0				
Mean	78/78	37.0 [33.0, 40.0]	696.5 [561.6, 863.0]	20.3 [15.6, 24.9]
Geometric	78/78	36.0 [32.0, 38.5]	680.5 [540.4, 846.7]	21.5 [16.3, 25.5]
Harmonic	78/78	35.5 [31.0, 38.0]	666.9 [530.8, 834.3]	22.4 [17.2, 26.7]
Median	78/78	21.0 [18.0, 24.0]	116.1 [76.5, 184.1]	11.7 [9.6, 16.8]
Minimum	78/78	11.5 [9.0, 15.0]	98.1 [55.8, 154.6]	22.2 [19.3, 24.5]
Borda	78/78	48.0 [45.5, 50.5]	549.1 [407.7, 714.4]	16.1 [12.2, 20.0]
Condorcet req.	78/78	31.0 [31.0, 32.0]	258.7 [198.0, 340.0]	11.9 [8.9, 15.2]
Plurality	78/78	18.0 [18.0, 18.0]	149.1 [111.8, 187.4]	10.3 [8.0, 13.0]
front2				
Mean	78/78	31.0 [27.0, 34.0]	696.5 [555.9, 852.4]	16.3 [12.0, 19.6]
Geometric	78/78	30.5 [26.5, 33.0]	676.8 [542.5, 844.3]	17.2 [12.9, 21.1]
Harmonic	78/78	30.0 [26.0, 33.0]	659.3 [520.7, 825.0]	18.9 [13.7, 22.2]
Median	78/78	21.0 [18.0, 23.5]	112.3 [71.0, 163.4]	9.8 [7.8, 12.7]
Minimum	78/78	11.5 [9.0, 15.0]	98.1 [55.8, 153.5]	22.2 [19.3, 25.0]
Borda	78/78	47.0 [44.0, 49.5]	533.6 [396.0, 695.9]	15.1 [11.3, 18.5]
Condorcet req.	78/78	29.0 [29.0, 29.0]	213.5 [161.5, 259.0]	10.2 [7.8, 13.0]
Plurality	78/78	9.0 [9.0, 9.0]	47.2 [33.8, 59.9]	6.2 [5.3, 7.8]
ceil				
Mean	78/78	16.0 [13.0, 18.5]	696.5 [559.9, 863.0]	13.2 [10.3, 16.5]
Geometric	78/78	16.0 [13.0, 18.5]	668.7 [535.2, 834.5]	14.2 [11.1, 17.5]
Harmonic	78/78	16.0 [13.0, 20.0]	643.6 [509.2, 808.0]	15.3 [12.0, 18.7]
Median	78/78	21.0 [18.0, 24.0]	112.3 [73.8, 163.4]	9.8 [7.6, 13.1]
Minimum	78/78	11.5 [9.0, 15.0]	98.1 [57.8, 153.8]	22.2 [19.3, 24.5]
Borda	78/78	47.0 [44.0, 49.0]	533.6 [393.0, 687.0]	15.1 [11.2, 18.5]
Condorcet req.	78/78	29.0 [29.0, 29.0]	213.5 [159.4, 271.2]	10.2 [7.8, 13.1]
Plurality	78/78	9.0 [9.0, 9.0]	47.2 [33.8, 59.9]	6.2 [5.3, 7.8]

Table 8: Ordinal/mean L_1 ratios: median [95% target-bootstrap interval] on jointly feasible targets. The Both column gives the joint denominator; Mean only counts targets feasible under the mean but infeasible under the ordinal requirement.

Dataset	Cap	Rule	Both	L_1 ratio	Mean only
HELM	gain5	Borda	10/78	0.4 [0.3, 0.7]	1
HELM	gain5	Condorcet req.	11/78	0.2 [0.093, 0.2]	0
HELM	gain5	Plurality	11/78	0.035 [0.021, 0.066]	0
HELM	gain10	Borda	25/78	0.6 [0.5, 0.7]	4
HELM	gain10	Condorcet req.	29/78	0.2 [0.2, 0.3]	0
HELM	gain10	Plurality	29/78	0.055 [0.047, 0.063]	0
HELM	gain20	Borda	49/78	0.7 [0.6, 0.7]	7
HELM	gain20	Condorcet req.	56/78	0.3 [0.3, 0.3]	0
HELM	gain20	Plurality	56/78	0.063 [0.058, 0.067]	0
HELM	front0	Borda	78/78	0.8 [0.8, 0.8]	0
HELM	front0	Condorcet req.	78/78	0.4 [0.3, 0.4]	0
HELM	front0	Plurality	78/78	0.2 [0.2, 0.2]	0
HELM	front2	Borda	78/78	0.8 [0.7, 0.8]	0
HELM	front2	Condorcet req.	78/78	0.3 [0.3, 0.3]	0
HELM	front2	Plurality	78/78	0.068 [0.063, 0.073]	0
HELM	ceil	Borda	78/78	0.8 [0.7, 0.8]	0
HELM	ceil	Condorcet req.	78/78	0.3 [0.3, 0.3]	0
HELM	ceil	Plurality	78/78	0.068 [0.063, 0.073]	0

Dataset	Cap	Rule	Both	L_1 ratio	Mean only
OLL sample	gain5	Borda	9/136	0.2 [0.094, 0.4]	4
OLL sample	gain5	Condorcet req.	5/136	0.8 [0.4, 1.4]	8
OLL sample	gain5	Plurality	3/136	0.9 [0.7, 1.1]	10
OLL sample	gain10	Borda	65/136	0.3 [0.3, 0.4]	2
OLL sample	gain10	Condorcet req.	43/136	0.5 [0.5, 0.6]	24
OLL sample	gain10	Plurality	13/136	0.5 [0.4, 0.5]	54
OLL sample	gain20	Borda	101/136	0.4 [0.4, 0.4]	4
OLL sample	gain20	Condorcet req.	96/136	0.6 [0.6, 0.7]	9
OLL sample	gain20	Plurality	89/136	0.5 [0.4, 0.5]	16
OLL sample	emp95	Borda	73/136	0.4 [0.3, 0.4]	12
OLL sample	emp95	Condorcet req.	20/136	0.5 [0.4, 0.5]	65
OLL sample	emp95	Plurality	40/136	0.6 [0.6, 0.7]	45
OLL sample	emp99	Borda	92/136	0.4 [0.4, 0.4]	2
OLL sample	emp99	Condorcet req.	56/136	0.6 [0.5, 0.6]	38
OLL sample	emp99	Plurality	59/136	0.6 [0.5, 0.6]	35
OLL sample	front0	Borda	136/136	0.5 [0.4, 0.5]	0
OLL sample	front0	Condorcet req.	136/136	0.7 [0.6, 0.7]	0
OLL sample	front0	Plurality	136/136	0.8 [0.7, 0.8]	0
OLL sample	front2	Borda	136/136	0.5 [0.4, 0.5]	0
OLL sample	front2	Condorcet req.	136/136	0.7 [0.6, 0.7]	0
OLL sample	front2	Plurality	136/136	0.4 [0.4, 0.5]	0
OLL sample	ceil	Borda	136/136	0.5 [0.4, 0.5]	0
OLL sample	ceil	Condorcet req.	136/136	0.7 [0.6, 0.7]	0
OLL sample	ceil	Plurality	136/136	0.4 [0.4, 0.5]	0

Table 9: Task-count inequalities hold only conditionally: counts on jointly feasible targets, with full infeasibility denominators in the display CSV.

Dataset	Cap	Proposed inequality	Holds
HELM	ceil	K(mean) $\leq$ K(median)	67/78
HELM	ceil	K(median) $\leq$ K(min)	7/78
HELM	front0	K(mean) $\leq$ K(median)	3/78
HELM	front0	K(median) $\leq$ K(min)	7/78
HELM	gain10	K(mean) $\leq$ K(median)	2/29
HELM	gain10	K(median) $\leq$ K(min)	0/13
OLLB census	ceil	K(mean) $\leq$ K(trimmed)	4495/4495
OLLB census	ceil	K(trimmed) $\leq$ K(median)	4494/4495
OLLB census	ceil	K(median) $\leq$ K(min)	4003/4495
OLLB census	front0	K(mean) $\leq$ K(trimmed)	3317/4495
OLLB census	front0	K(trimmed) $\leq$ K(median)	2757/4495
OLLB census	front0	K(median) $\leq$ K(min)	3694/4495
OLLB census	gain10	K(mean) $\leq$ K(trimmed)	18/50
OLLB census	gain10	K(trimmed) $\leq$ K(median)	3/53
OLLB census	gain10	K(median) $\leq$ K(min)	33/113
OLLB matched sample	ceil	K(mean) $\leq$ K(trimmed)	136/136
OLLB matched sample	ceil	K(trimmed) $\leq$ K(median)	135/136
OLLB matched sample	ceil	K(median) $\leq$ K(min)	120/136
OLLB matched sample	front0	K(mean) $\leq$ K(trimmed)	93/136
OLLB matched sample	front0	K(trimmed) $\leq$ K(median)	73/136
OLLB matched sample	front0	K(median) $\leq$ K(min)	118/136
OLLB matched sample	gain10	K(mean) $\leq$ K(trimmed)	18/43
OLLB matched sample	gain10	K(trimmed) $\leq$ K(median)	3/43
OLLB matched sample	gain10	K(median) $\leq$ K(min)	28/44

Table 10: Additional power means on OLLB: feasible counts and conditional medians with target interquartile ranges.

Cap	Exponent p	Feasible	K [Q25, Q75]	L_1 [Q25, Q75]
ceil	-0.5	4495/4495	3.0 [2.0, 4.0]	138.8 [101.3, 185.8]
ceil	-2	4495/4495	3.0 [3.0, 4.0]	100.3 [68.4, 144.3]
ceil	-5	4495/4495	3.0 [3.0, 5.0]	67.5 [43.6, 105.6]
ceil	0.5	4495/4495	3.0 [2.0, 3.0]	167.1 [125.0, 215.3]
gain10	-0.5	195/4495	5.0 [4.0, 6.0]	45.3 [34.0, 52.6]
gain10	-2	344/4495	2.0 [2.0, 4.0]	19.7 [16.6, 30.0]
gain10	-5	389/4495	2.0 [2.0, 2.0]	13.6 [11.1, 17.8]
gain10	0.5	112/4495	5.0 [4.0, 6.0]	46.8 [35.6, 55.9]
gain20	-0.5	1041/4495	4.0 [3.0, 5.0]	66.0 [40.0, 88.4]
gain20	-2	1668/4495	3.0 [2.0, 5.0]	53.6 [30.2, 86.1]
gain20	-5	2454/4495	3.0 [2.0, 4.0]	48.1 [28.9, 61.9]
gain20	0.5	881/4495	4.0 [3.0, 5.0]	78.6 [59.7, 99.1]

Table 11: Additional power means on HELM: feasible counts and conditional medians with target interquartile ranges.

Cap	Exponent p	Feasible	K [Q25, Q75]	L_1 [Q25, Q75]
ceil	-0.5	78/78	16.0 [10.2, 22.8]	656.3 [413.5, 1014.4]
ceil	-2	78/78	16.0 [10.2, 23.8]	616.7 [381.7, 971.7]
ceil	-5	78/78	15.5 [10.2, 23.8]	527.3 [310.5, 866.4]
ceil	0.5	78/78	16.0 [10.0, 22.8]	682.2 [432.3, 1041.6]
gain10	-0.5	22/78	29.0 [17.2, 44.5]	281.9 [168.9, 430.3]
gain10	-2	18/78	27.5 [15.0, 38.0]	271.6 [140.9, 374.5]
gain10	-5	13/78	21.0 [9.0, 26.0]	201.0 [82.2, 248.1]
gain10	0.5	26/78	35.0 [19.0, 46.0]	342.1 [184.8, 455.6]
gain20	-0.5	50/78	26.5 [16.2, 34.0]	498.9 [312.0, 647.7]
gain20	-2	49/78	25.0 [15.0, 35.0]	472.0 [279.2, 660.9]
gain20	-5	40/78	20.0 [10.0, 26.0]	373.6 [181.2, 470.5]
gain20	0.5	53/78	27.0 [17.0, 38.0]	527.2 [333.0, 731.1]

Table 12: Reported costs for the excluded original mean leader on OLLB, kept separate from all nonleader summaries. Each row is one target, with separately minimized costs.

Rule	K	L_1	L_∞
gain5			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	2.0	8.3	4.1
Minimum	1.0	2.8	2.8
gain10			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	1.0	8.3	4.1
Minimum	1.0	2.8	2.8

Rule	K	L_1	L_∞
gain20			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	1.0	8.3	4.1
Minimum	1.0	2.8	2.8
emp95			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	1.0	8.3	4.1
Minimum	1.0	2.8	2.8
emp99			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	1.0	8.3	4.1
Minimum	1.0	2.8	2.8
front0			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	1.0	8.3	4.1
Minimum	1.0	2.8	2.8
front2			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	1.0	8.3	4.1
Minimum	1.0	2.8	2.8
ceil			
Mean	0	0	0
Geometric	0	0	0
Harmonic	0	0	0
Trimmed	0	0	0
Median	1.0	8.3	4.1
Minimum	1.0	2.8	2.8

Table 13: Reported costs for the excluded original mean leader on HELM, kept separate from all nonleader summaries. Each row is one target, with separately minimized costs.

Rule	K	L_1	L_∞
gain5			
Mean	0	0	0
Geometric	1.0	1.1	0.027
Harmonic	1.0	4.1	0.1
Median	1.0	0.001	0.001
Minimum	1.0	1.2	1.2
Borda	0	0	0
Condorcet req.	0	0	0
Plurality	2.0	0.003	0.001

Rule	K	L_1	L_∞
gain10			
Mean	0	0	0
Geometric	1.0	1.1	0.027
Harmonic	1.0	4.1	0.1
Median	1.0	0.001	0.001
Minimum	1.0	1.2	1.2
Borda	0	0	0
Condorcet req.	0	0	0
Plurality	2.0	0.003	0.001
gain20			
Mean	0	0	0
Geometric	1.0	1.1	0.027
Harmonic	1.0	4.1	0.1
Median	1.0	0.001	0.001
Minimum	1.0	1.2	1.2
Borda	0	0	0
Condorcet req.	0	0	0
Plurality	2.0	0.003	0.001
front0			
Mean	0	0	0
Geometric	1.0	1.1	0.035
Harmonic	1.0	4.3	0.2
Median	1.0	0.001	0.001
Minimum	1.0	1.2	1.2
Borda	0	0	0
Condorcet req.	0	0	0
Plurality	3.0	2.0	0.7
front2			
Mean	0	0	0
Geometric	1.0	1.1	0.028
Harmonic	1.0	4.1	0.1
Median	1.0	0.001	0.001
Minimum	1.0	1.2	1.2
Borda	0	0	0
Condorcet req.	0	0	0
Plurality	2.0	0.003	0.001
ceil			
Mean	0	0	0
Geometric	1.0	1.1	0.027
Harmonic	1.0	4.1	0.1
Median	1.0	0.001	0.001
Minimum	1.0	1.2	1.2
Borda	0	0	0
Condorcet req.	0	0	0
Plurality	2.0	0.003	0.001

Table 14: Data slices, score normalization and duplicate selection: feasible/total counts and conditional L_1 medians with 95% target-bootstrap intervals.

Slice	Cap	Rule	Feasible	L_1 median [95% CI]
Normalised	ceil	Geometric	4495/4495	153.2 [150.7, 154.7]
Normalised	ceil	Harmonic	4495/4495	124.8 [123.6, 126.5]
Normalised	ceil	Mean	4495/4495	180.5 [177.8, 182.3]
Normalised	ceil	Median	4495/4495	105.9 [104.8, 107.5]
Normalised	ceil	Minimum	4495/4495	43.2 [42.5, 44.1]
Normalised	front0	Geometric	4495/4495	160.7 [158.2, 162.6]
Normalised	front0	Harmonic	4495/4495	132.6 [130.7, 134.0]
Normalised	front0	Mean	4495/4495	180.5 [177.9, 182.3]
Normalised	front0	Median	4495/4495	109.1 [107.5, 110.4]
Normalised	front0	Minimum	4495/4495	43.2 [42.5, 44.1]

Slice	Cap	Rule	Feasible	L_1 median [95% CI]
Normalised	gain10	Geometric	142/4495	48.4 [44.7, 49.9]
Normalised	gain10	Harmonic	264/4495	39.5 [37.8, 41.2]
Normalised	gain10	Mean	113/4495	50.9 [47.5, 52.6]
Normalised	gain10	Median	203/4495	18.3 [16.6, 20.5]
Normalised	gain10	Minimum	446/4495	10.4 [9.8, 11.0]
No merges	ceil	Geometric	3788/3788	156.1 [153.9, 158.1]
No merges	ceil	Harmonic	3788/3788	127.6 [126.0, 129.2]
No merges	ceil	Mean	3788/3788	183.3 [181.5, 185.8]
No merges	ceil	Median	3788/3788	108.0 [106.2, 109.6]
No merges	ceil	Minimum	3788/3788	42.4 [41.3, 43.3]
No merges	front0	Geometric	3788/3788	166.4 [164.7, 168.9]
No merges	front0	Harmonic	3788/3788	140.9 [138.7, 142.9]
No merges	front0	Mean	3788/3788	183.3 [181.4, 185.8]
No merges	front0	Median	3788/3788	110.6 [109.1, 112.6]
No merges	front0	Minimum	3788/3788	42.4 [41.4, 43.3]
No merges	gain10	Geometric	106/3788	48.1 [43.4, 50.2]
No merges	gain10	Harmonic	196/3788	38.9 [36.5, 41.2]
No merges	gain10	Mean	81/3788	47.9 [41.0, 52.2]
No merges	gain10	Median	150/3788	18.1 [16.4, 20.1]
No merges	gain10	Minimum	389/3788	9.6 [8.7, 10.4]
Top 500	ceil	Geometric	499/499	48.5 [47.8, 49.7]
Top 500	ceil	Harmonic	499/499	31.9 [31.2, 32.7]
Top 500	ceil	Mean	499/499	71.1 [69.3, 73.2]
Top 500	ceil	Median	499/499	26.5 [25.5, 27.5]
Top 500	ceil	Minimum	499/499	12.3 [11.4, 13.2]
Top 500	front0	Geometric	499/499	52.7 [51.5, 54.5]
Top 500	front0	Harmonic	499/499	33.0 [31.8, 33.7]
Top 500	front0	Mean	499/499	71.1 [69.3, 73.1]
Top 500	front0	Median	499/499	26.7 [25.5, 27.6]
Top 500	front0	Minimum	499/499	12.3 [11.4, 13.3]
Top 500	gain10	Geometric	142/499	48.4 [44.8, 49.9]
Top 500	gain10	Harmonic	261/499	39.4 [37.6, 41.0]
Top 500	gain10	Mean	113/499	50.9 [47.5, 52.8]
Top 500	gain10	Median	200/499	18.3 [16.5, 20.3]
Top 500	gain10	Minimum	348/499	9.6 [9.2, 10.1]
Raw	ceil	Geometric	4495/4495	140.6 [138.2, 142.4]
Raw	ceil	Harmonic	4495/4495	126.6 [124.4, 128.2]
Raw	ceil	Mean	4495/4495	154.1 [151.7, 155.8]
Raw	ceil	Median	4495/4495	92.9 [91.3, 94.3]
Raw	ceil	Minimum	4495/4495	74.7 [73.7, 75.6]
Raw	front0	Geometric	4495/4495	142.3 [139.9, 143.9]
Raw	front0	Harmonic	4495/4495	129.0 [126.7, 130.6]
Raw	front0	Mean	4495/4495	154.1 [151.7, 155.7]
Raw	front0	Median	4495/4495	124.3 [122.4, 125.9]
Raw	front0	Minimum	4495/4495	74.7 [73.7, 75.6]
Raw	gain10	Geometric	382/4495	42.1 [41.0, 43.9]
Raw	gain10	Harmonic	459/4495	33.7 [32.3, 35.4]
Raw	gain10	Mean	296/4495	48.5 [46.9, 49.9]
Raw	gain10	Median	21/4495	15.7 [14.8, 17.5]
Raw	gain10	Minimum	224/4495	12.6 [12.1, 13.1]
Dedupe worst	ceil	Geometric	4495/4495	153.5 [151.1, 155.0]
Dedupe worst	ceil	Harmonic	4495/4495	125.0 [123.8, 126.6]
Dedupe worst	ceil	Mean	4495/4495	180.7 [178.2, 182.5]
Dedupe worst	ceil	Median	4495/4495	106.1 [105.0, 107.5]
Dedupe worst	ceil	Minimum	4495/4495	43.2 [42.5, 44.2]
Dedupe worst	front0	Geometric	4495/4495	161.1 [158.4, 162.8]
Dedupe worst	front0	Harmonic	4495/4495	132.8 [131.0, 134.1]
Dedupe worst	front0	Mean	4495/4495	180.7 [178.2, 182.4]
Dedupe worst	front0	Median	4495/4495	109.3 [107.5, 110.5]
Dedupe worst	front0	Minimum	4495/4495	43.2 [42.5, 44.1]
Dedupe worst	gain10	Geometric	142/4495	48.7 [45.8, 50.3]
Dedupe worst	gain10	Harmonic	259/4495	39.4 [37.4, 41.2]
Dedupe worst	gain10	Mean	111/4495	50.8 [47.2, 52.6]
Dedupe worst	gain10	Median	201/4495	18.3 [16.6, 20.4]
Dedupe worst	gain10	Minimum	444/4495	10.5 [9.8, 11.1]

Table 15: Ceiling-cap offsets sensitivity of conditional median L_1, with feasibility shares.

Dataset	Rule	Parameter	Feasible	Median L_1
HELM	Geometric	0.1	100%	668.4
HELM	Geometric	1.0	100%	668.7
HELM	Geometric	10.0	100%	671.1
HELM	Harmonic	0.1	100%	643.0
HELM	Harmonic	1.0	100%	643.6
HELM	Harmonic	10.0	100%	648.7
OLLB	Geometric	0.1	100%	152.6
OLLB	Geometric	1.0	100%	153.2
OLLB	Geometric	10.0	100%	157.9
OLLB	Harmonic	0.1	100%	123.5
OLLB	Harmonic	1.0	100%	124.8
OLLB	Harmonic	10.0	100%	135.1

Table 16: Ceiling-cap eps sensitivity of conditional median L_1, with feasibility shares.

Dataset	Rule	Parameter	Feasible	Median L_1
HELM	Borda	0.0001	100%	533.5
HELM	Condorcet req.	0.0001	100%	213.4
HELM	Geometric	0.0001	100%	668.7
HELM	Harmonic	0.0001	100%	643.5
HELM	Mean	0.0001	100%	696.4
HELM	Median	0.0001	100%	112.3
HELM	Minimum	0.0001	100%	98.1
HELM	Plurality	0.0001	100%	47.2
HELM	Borda	0.01	100%	534.1
HELM	Condorcet req.	0.01	100%	213.7
HELM	Geometric	0.01	100%	669.2
HELM	Harmonic	0.01	100%	644.1
HELM	Mean	0.01	100%	697.0
HELM	Median	0.01	100%	112.5
HELM	Minimum	0.01	100%	98.2
HELM	Plurality	0.01	100%	47.3
HELM	Borda	0.1	100%	539.1
HELM	Condorcet req.	0.1	100%	216.4
HELM	Geometric	0.1	100%	674.3
HELM	Harmonic	0.1	100%	649.0
HELM	Mean	0.1	100%	702.1
HELM	Median	0.1	100%	114.4
HELM	Minimum	0.1	100%	99.4
HELM	Plurality	0.1	100%	48.1
OLLB	Geometric	0.0001	100%	153.2
OLLB	Harmonic	0.0001	100%	124.8
OLLB	Mean	0.0001	100%	180.5
OLLB	Median	0.0001	100%	105.9
OLLB	Minimum	0.0001	100%	43.2
OLLB	Geometric	0.01	100%	153.2
OLLB	Harmonic	0.01	100%	124.9
OLLB	Mean	0.01	100%	180.6
OLLB	Median	0.01	100%	105.9
OLLB	Minimum	0.01	100%	43.2
OLLB	Geometric	0.1	100%	153.8
OLLB	Harmonic	0.1	100%	125.4
OLLB	Mean	0.1	100%	181.1
OLLB	Median	0.1	100%	106.3
OLLB	Minimum	0.1	100%	43.5
HELM	Borda	0.001	100%	533.6
HELM	Condorcet req.	0.001	100%	213.5
HELM	Geometric	0.001	100%	668.7

Dataset	Rule	Parameter	Feasible	Median L_1
HELM	Harmonic	0.001	100%	643.6
HELM	Mean	0.001	100%	696.5
HELM	Median	0.001	100%	112.3
HELM	Minimum	0.001	100%	98.1
HELM	Plurality	0.001	100%	47.2
OLLB	Geometric	0.001	100%	153.2
OLLB	Harmonic	0.001	100%	124.8
OLLB	Mean	0.001	100%	180.5
OLLB	Median	0.001	100%	105.9
OLLB	Minimum	0.001	100%	43.2

Table 17: Reoptimized cost distributions for original mean runner-ups under gain10: draw quantiles, not fixed-intervention success probabilities.

Dataset / resampling	Rule	Nominal L_1	Feas. draws	L_1 [5th, 50th, 95th]
HELM / noise	Borda	5.6	100%	0 / 7.8 / 27.1
HELM / noise	Condorcet req.	0.002	100%	0 / 1.3 / 5.9
HELM / noise	Geometric	0	100%	0 / 0 / 37.4
HELM / noise	Mean	4.0	100%	0 / 4.0 / 44.8
HELM / noise	Median	0.001	100%	0 / 1.1 / 5.9
HELM / noise	Minimum	0	100%	0 / 1.7 / 7.5
HELM / noise	Plurality	1.4	100%	0 / 0.9 / 4.4
OLLB / noise	Geometric	2.9	100%	0 / 3.1 / 10.8
OLLB / noise	Mean	4.8	100%	0 / 4.9 / 15.1
OLLB / noise	Median	9.6	100%	6.9 / 10.1 / 13.4
OLLB / noise	Minimum	3.7	95%	3.1 / 5.9 / 8.8
HELM / taskboot	Borda	5.6	100%	0.008 / 5.9 / 17.5
HELM / taskboot	Geometric	0	100%	0 / 1.6 / 40.4
HELM / taskboot	Mean	4.0	100%	0 / 5.2 / 50.5
HELM / taskboot	Median	0.001	100%	0 / 0.9 / 5.3
HELM / taskboot	Minimum	0	100%	0 / 0 / 4.0
HELM / taskboot	Plurality	1.4	100%	0.004 / 2.1 / 5.7
OLLB / taskboot	Geometric	2.9	97%	1.6 / 4.5 / 25.1
OLLB / taskboot	Mean	4.8	97%	2.3 / 6.0 / 33.7
OLLB / taskboot	Median	9.6	82%	0.4 / 3.3 / 16.8
OLLB / taskboot	Minimum	3.7	94%	2.0 / 6.6 / 19.9

Table 18: Weighted-mean sensitivity: distributions across 50 Dirichlet(1) weight draws, with equal-weight costs as reference.

Dataset	Cap	Mean rank	Feas.	Equal L_1	Weighted L_1 [5, 50, 95]
HELM	ceil	5	100%	154.0	31.5 / 45.1 / 63.2
HELM	ceil	2	100%	4.0	0 / 1.6 / 11.9
HELM	ceil	10	100%	260.2	52.4 / 81.6 / 130.8
HELM	ceil	3	100%	22.5	0 / 7.0 / 17.6
HELM	ceil	20	100%	429.5	105.7 / 142.1 / 198.4
HELM	ceil	50	100%	886.3	284.8 / 361.7 / 466.6
HELM	gain10	5	100%	154.0	33.9 / 52.1 / 84.5
HELM	gain10	2	100%	4.0	0 / 1.6 / 12.0
HELM	gain10	10	100%	260.2	62.0 / 105.6 / 196.3
HELM	gain10	3	100%	22.5	0 / 7.0 / 18.6
HELM	gain10	20	100%	429.5	169.0 / 219.0 / 375.0
HELM	gain10	50	0%	∞	– / – / –
OLLB	ceil	50	100%	48.0	11.8 / 24.5 / 36.8
OLLB	ceil	2	100%	4.8	1.7 / 3.0 / 10.7
OLLB	ceil	10	100%	28.5	2.0 / 16.5 / 26.9
OLLB	ceil	3	100%	5.1	1.8 / 3.5 / 10.1
OLLB	ceil	5	100%	23.9	0.2 / 14.1 / 31.0
OLLB	ceil	100	100%	58.0	12.8 / 29.1 / 42.4

Dataset	Cap	Mean rank	Feas.	Equal L_1	Weighted L_1 [5, 50, 95]
OLLB	ceil	20	100%	32.1	7.1 / 17.5 / 26.1
OLLB	gain10	50	58%	48.0	13.2 / 24.2 / 41.0
OLLB	gain10	2	100%	4.8	1.7 / 3.0 / 11.1
OLLB	gain10	10	84%	28.5	1.8 / 13.4 / 45.8
OLLB	gain10	3	100%	5.1	1.8 / 3.5 / 10.3
OLLB	gain10	5	78%	23.9	0.1 / 6.5 / 34.3
OLLB	gain10	100	44%	58.0	12.9 / 24.2 / 46.8
OLLB	gain10	20	90%	32.1	7.0 / 20.5 / 40.8

D ADDITIONAL ANALYSES ADDED AFTER EXTERNAL REVIEW

Definitions. *Leader gap*: the best rival's rule score minus the target's rule score (zero if the target already leads); ***IQR***: the interquartile range of the rule score over all models; normalised costs are L_1/gap and (L_1/n)/IQR, defined only for rules with a score in task-point units (mean, geometric, harmonic, median, minimum, trimmed mean). *Rank-matched*: the ratio of a rule's median cost to the mean's among targets whose rank under that rule falls in the same bin. *Replay*: uniform raise of every task by d, capped, against $R = 200$ noise draws shared across rules, targets and budgets; p_0 is the share of draws in which the target is strictly first at the nominal L_∞, and x_{90} the further raise until that share reaches 0.9 (censored beyond 16 points). *Shrunk centring*: noise is drawn around $T = m + \lambda(X - m)$ with m the column mean and $\lambda = \max(0, 1 - \overline{\mathrm{SE}^2}/\mathrm{Var}(X))$, with homoscedastic Gaussian noise and one prior mean per task. *Cluster*: the `Base Model` field of the OLLB table; a model whose base is a placeholder or itself is its own cluster. Intervals are percentile bootstrap over targets (B=2000, seed 20261108) unless the caption says Wilson, draw quantile or cluster bootstrap.

Sizes of the noise runs. The plan fixed 200 draws per run; some runs are smaller, and none of them is a sample of the whole leaderboard. Replay (Section 4.5): 200 draws, 36 targets per dataset; Figures 8 and 9 use the earlier 6 targets (ranks 1, 2, 3, 5, 10, 20). Re-optimised noise runs at gain10: 100 draws, the 10 best-ranked non-leaders on HELM and the 20 best-ranked on OLLB, not a stratified sample; re-run for the score rules with 200 draws and the study seed, the median costs move by at most 1.0 (OLLB) and 6.6 points (HELM), inside the bootstrap intervals. HELM ordinal rules were not re-run with 200 draws since each draw costs about 155 s of exact optimisation. Bootstrap over tasks: 100 draws on HELM and 200 on OLLB. Task-subset and random-weight runs: 50 draws. The first noise and bootstrap runs used the script's default seed 0 (Section 5).

Number of tasks and cross-task correlation. On HELM we draw 50 random subsets of $n \in \{3, 6, 12, 24, 48, 57\}$ subjects and recompute the cost for a target of rank 5 within the subset, for the observed matrix and for a control in which each task column is permuted over models, which removes cross-task correlation (Figure 4a; the L_1/n and K/n curves for ranks 2 and 5 are in Figure 10). In the observed matrix the mean's L_1 rises from 6.1 at $n = 3$ to 154.0 at $n = 57$, roughly in proportion to n, as the closed form predicts for a gap that persists across tasks. In the shuffled matrix it rises only to 49.4 at $n = 57$: with correlation removed the average over many tasks concentrates and a target's gap shrinks with n. The median behaves in the opposite way: its L_1 falls with n in the observed matrix (4.3 at $n = 12$, 1.5 at $n = 57$). The number of tasks K grows slowly (mean: 1.0 at $n = 3$ to 4.0 at $n = 57$), so K/n falls. We read these as a comparison of curves with and without correlation, not as fitted exponents, and as a within-benchmark proxy for n, since subsets of MMLU subjects are not independent leaderboards.

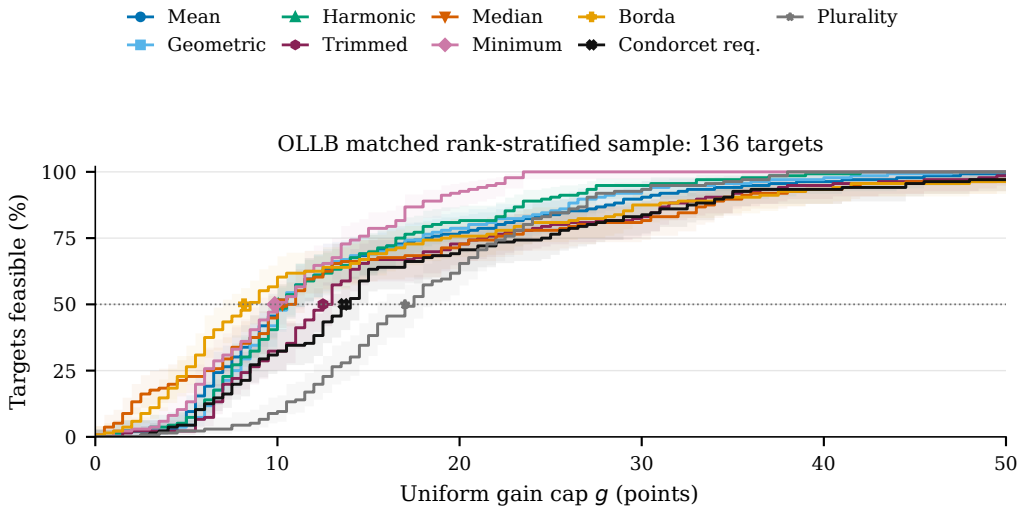


Figure 12: Cap-threshold curves on the same 136 OLLB targets for all nine rules. This deterministic rank-stratified sample over-represents top ranks and is separate from the census. Markers give exact lower-median crossings; light shading gives pointwise Wilson 95% reference intervals across targets. The display range is 0–50 points.

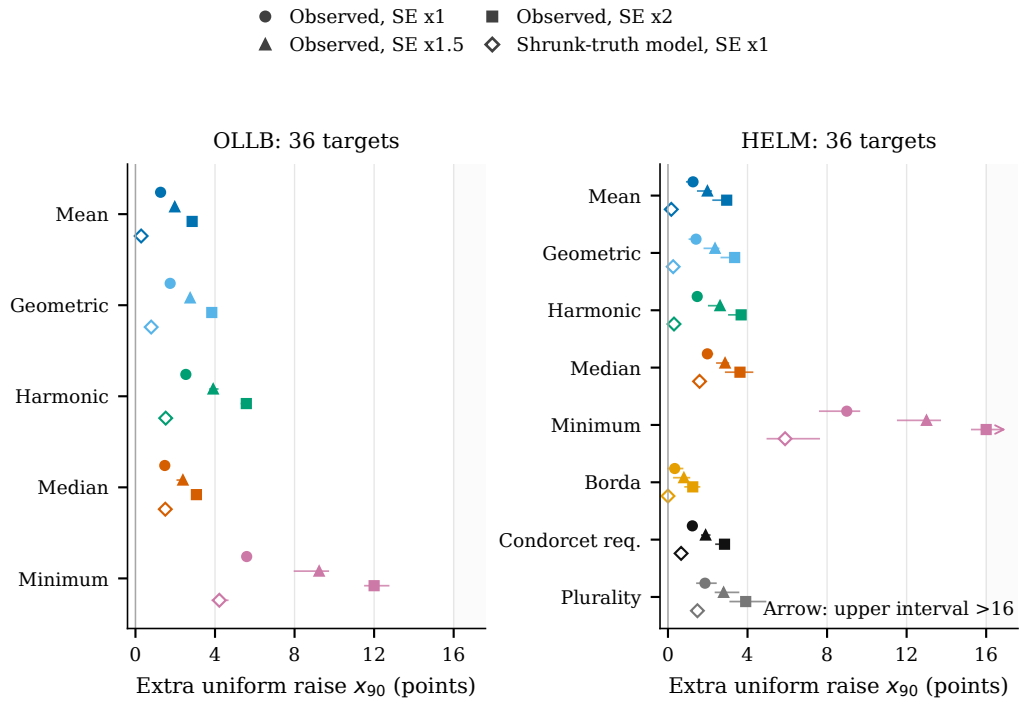


Figure 13: Extra per-task gain beyond the nominal L_∞ optimum needed for 90% replay success. Points are medians over 36 targets per board; bars are 95% target-bootstrap intervals. Noise multipliers and the open-diamond shrunk-truth centre are model sensitivities. A right arrow retains censoring beyond the evaluated +16 grid.

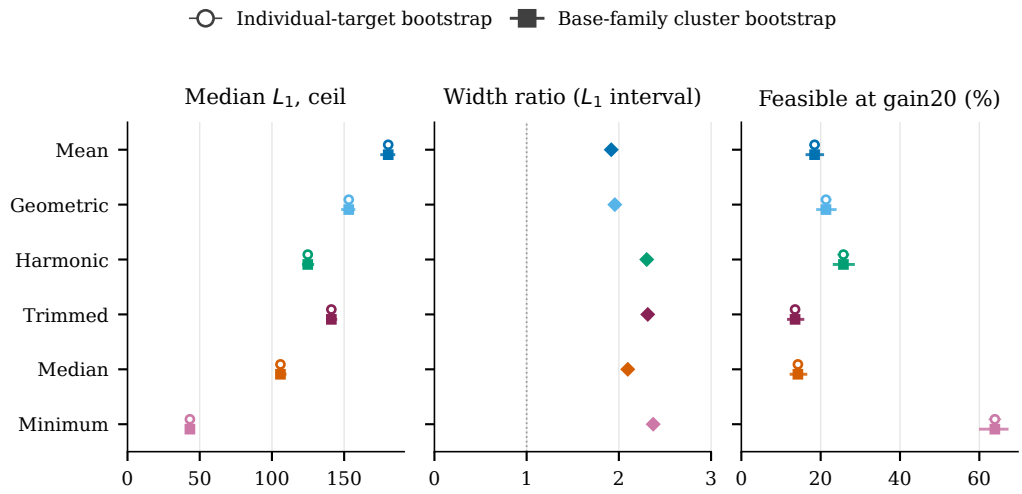


Figure 14: Base-family dependence widens OLLB intervals while preserving point estimates. Open circles resample individual targets; filled squares resample 3849 base-model families (4495 targets). Bars are 95% percentile intervals. Middle diamonds give the cluster/individual interval-width ratio for median L_1; these are display ratios, not extra fitted estimates.

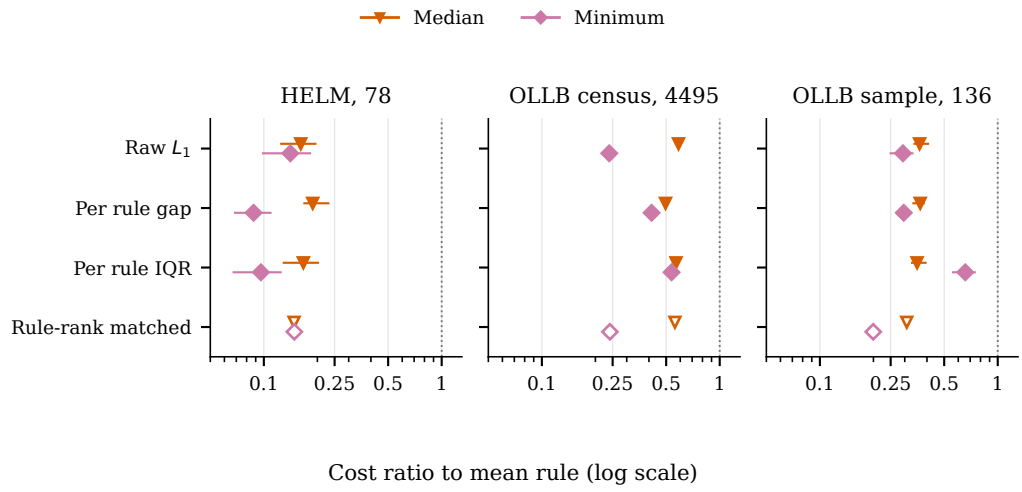


Figure 15: Median/minimum L_1 advantage depends on the leaderboard and score currency. Ratios below one favor the named rule over the mean. Filled points have 95% paired-target-bootstrap intervals; open rule-rank-matched points are geometric summaries of bin ratios, with no supplied aggregate interval.

Table 19: Cap needed for at least 50% feasibility: the lower median of g^*, in per-task score points, with 95% empirical-target-bootstrap intervals. Feasible shares at gain5/10/20 use the same targets; all thresholds are finite at ceil.

Rule	g_{50} [95% interval]	Feas. $g = 5$	Feas. $g = 10$	Feas. $g = 20$
OLLB census, 4495 targets				
Mean	30.1 [29.7, 30.4]	0.3%	2.5%	18.5%
Geometric	27.8 [27.5, 28.0]	0.1%	3.2%	21.3%
Harmonic	24.9 [24.6, 25.1]	0.3%	5.9%	25.7%
Trimmed	33.7 [33.3, 34.0]	0.1%	1.2%	13.5%
Median	36.2 [35.8, 36.5]	0.9%	4.5%	14.3%
Minimum	17.9 [17.7, 18.1]	2.3%	9.9%	63.9%
HELM, 78 targets				
Mean	12.6 [10.2, 16.4]	14.1%	37.2%	71.8%
Geometric	13.8 [11.0, 17.5]	10.3%	30.8%	64.1%
Harmonic	15.2 [12.0, 18.7]	10.3%	25.6%	64.1%
Median	9.8 [7.5, 12.7]	20.5%	51.3%	84.6%
Minimum	22.2 [19.3, 24.5]	9.0%	17.9%	43.6%
Borda	15.0 [11.0, 17.9]	14.1%	32.1%	62.8%
Condorcet req.	10.1 [7.7, 12.7]	20.5%	48.7%	83.3%
Plurality	6.1 [5.3, 7.5]	37.2%	73.1%	94.9%
OLLB matched sample, 136 targets				
Mean	10.1 [8.9, 11.1]	9.6%	49.3%	77.2%
Geometric	10.3 [9.3, 11.3]	4.4%	47.8%	78.7%
Harmonic	10.0 [9.6, 11.3]	7.4%	48.5%	81.6%
Trimmed	12.5 [11.0, 13.3]	2.2%	32.4%	72.8%
Median	10.3 [9.1, 11.2]	22.8%	49.3%	72.8%
Minimum	9.9 [8.7, 11.1]	13.2%	50.7%	92.6%
Borda	8.2 [6.7, 9.7]	26.5%	60.3%	75.7%
Condorcet req.	13.6 [12.3, 14.4]	4.4%	32.4%	70.6%
Plurality	17.0 [15.3, 18.2]	2.2%	9.6%	65.4%

Table 20: Observed fine-tune-minus-base gain quantiles on OLLB, in leaderboard points (540 pairs naming an on-board base). This selected observational subset is a scale anchor, not an estimate of what targeted training buys.

Task	Q50	Q95	Q99	Maximum
IFEval	8.9	46.0	63.0	66.1
BBH	-0.1	7.3	12.1	34.9
MATH L5	0.4	21.3	25.7	33.0
GPQA	-0.2	3.8	6.4	10.0
MuSR	-0.8	7.2	12.1	22.8
MMLU-Pro	-2.3	3.2	7.1	30.7

Table 21: Expanded replay sensitivity. p_0 is success at the nominal uniform optimum; x_{90} is the further per-task raise for 90% success. Brackets bootstrap targets; Censored counts targets not reaching 90% by $+16$ points, retained in the summaries. The shrunk centre is a model.

Rule	Noise / centre	p_0 [95% interval]	x_{90} [95% interval]	Censored
OLLB, 36 targets; 200 draws per target				
Mean	Observed $\times 1$	0.43 [0.42, 0.45]	1.3 [1.2, 1.3]	0/36
Mean	Observed $\times 1.5$	0.38 [0.37, 0.40]	2.0 [1.9, 2.1]	0/36
Mean	Observed $\times 2$	0.34 [0.33, 0.36]	2.8 [2.7, 2.9]	0/36
Mean	Shrunk model $\times 1$	0.82 [0.79, 0.84]	0.3 [0.2, 0.4]	0/36
Geometric	Observed $\times 1$	0.40 [0.38, 0.41]	1.7 [1.7, 1.8]	0/36
Geometric	Observed $\times 1.5$	0.35 [0.34, 0.36]	2.8 [2.7, 2.8]	0/36
Geometric	Observed $\times 2$	0.31 [0.30, 0.33]	3.8 [3.7, 3.9]	0/36

Rule	Noise / centre	p_0 [95% interval]	x_{90} [95% interval]	Censored
Geometric	Shrunk model $\times 1$	0.72 [0.69, 0.80]	0.8 [0.5, 0.9]	0/36
Harmonic	Observed $\times 1$	0.34 [0.30, 0.34]	2.5 [2.5, 2.6]	0/36
Harmonic	Observed $\times 1.5$	0.28 [0.24, 0.30]	3.9 [3.8, 4.1]	0/36
Harmonic	Observed $\times 2$	0.24 [0.21, 0.27]	5.6 [5.3, 5.7]	0/36
Harmonic	Shrunk model $\times 1$	0.60 [0.55, 0.69]	1.5 [1.3, 1.7]	0/36
Median	Observed $\times 1$	0.26 [0.23, 0.27]	1.5 [1.4, 1.5]	0/36
Median	Observed $\times 1.5$	0.23 [0.21, 0.25]	2.4 [2.1, 2.5]	0/36
Median	Observed $\times 2$	0.21 [0.17, 0.23]	3.1 [2.9, 3.2]	0/36
Median	Shrunk model $\times 1$	0.26 [0.24, 0.27]	1.5 [1.4, 1.6]	0/36
Minimum	Observed $\times 1$	0.14 [0.12, 0.17]	5.6 [5.4, 5.7]	0/36
Minimum	Observed $\times 1.5$	0.09 [0.07, 0.10]	9.2 [8.0, 9.7]	0/36
Minimum	Observed $\times 2$	0.06 [0.04, 0.07]	12.0 [11.5, 12.7]	0/36
Minimum	Shrunk model $\times 1$	0.30 [0.26, 0.39]	4.2 [3.9, 4.6]	0/36
HELM, 36 targets; 200 draws per target				
Mean	Observed $\times 1$	0.24 [0.20, 0.28]	1.3 [0.9, 1.3]	0/36
Mean	Observed $\times 1.5$	0.19 [0.16, 0.23]	2.0 [1.5, 2.2]	0/36
Mean	Observed $\times 2$	0.17 [0.14, 0.20]	3.0 [2.3, 3.1]	0/36
Mean	Shrunk model $\times 1$	0.84 [0.74, 0.91]	0.2 [0, 0.3]	0/36
Geometric	Observed $\times 1$	0.21 [0.19, 0.25]	1.4 [1.1, 1.5]	0/36
Geometric	Observed $\times 1.5$	0.18 [0.15, 0.21]	2.4 [1.8, 2.5]	0/36
Geometric	Observed $\times 2$	0.15 [0.13, 0.19]	3.4 [2.7, 3.6]	0/36
Geometric	Shrunk model $\times 1$	0.81 [0.69, 0.89]	0.3 [0.083, 0.4]	0/36
Harmonic	Observed $\times 1$	0.24 [0.20, 0.26]	1.5 [1.3, 1.6]	0/36
Harmonic	Observed $\times 1.5$	0.20 [0.17, 0.23]	2.6 [2.1, 2.7]	0/36
Harmonic	Observed $\times 2$	0.16 [0.14, 0.20]	3.7 [3.1, 3.8]	0/36
Harmonic	Shrunk model $\times 1$	0.81 [0.72, 0.86]	0.3 [0.1, 0.5]	0/36
Median	Observed $\times 1$	0.22 [0.18, 0.28]	2.0 [1.8, 2.2]	0/36
Median	Observed $\times 1.5$	0.18 [0.14, 0.23]	2.9 [2.5, 3.0]	0/36
Median	Observed $\times 2$	0.15 [0.12, 0.21]	3.6 [2.9, 4.2]	0/36
Median	Shrunk model $\times 1$	0.37 [0.30, 0.48]	1.6 [1.4, 1.7]	0/36
Minimum	Observed $\times 1$	0.24 [0.19, 0.28]	9.0 [7.6, 9.6]	0/36
Minimum	Observed $\times 1.5$	0.18 [0.15, 0.21]	13.0 [11.6, 13.7]	2/36
Minimum	Observed $\times 2$	0.15 [0.13, 0.17]	16.0 [15.3, >16]	17/36
Minimum	Shrunk model $\times 1$	0.46 [0.39, 0.55]	5.9 [5.0, 7.6]	0/36
Borda	Observed $\times 1$	0.80 [0.65, 0.92]	0.3 [0, 0.7]	0/36
Borda	Observed $\times 1.5$	0.72 [0.59, 0.87]	0.8 [0.3, 1.1]	0/36
Borda	Observed $\times 2$	0.62 [0.52, 0.79]	1.2 [0.9, 1.6]	0/36
Borda	Shrunk model $\times 1$	0.97 [0.87, 0.99]	0 [0, 0.071]	0/36
Condorcet req.	Observed $\times 1$	0.29 [0.20, 0.35]	1.2 [1.0, 1.4]	0/36
Condorcet req.	Observed $\times 1.5$	0.19 [0.17, 0.23]	1.9 [1.7, 2.1]	0/36
Condorcet req.	Observed $\times 2$	0.14 [0.12, 0.15]	2.8 [2.4, 3.0]	0/36
Condorcet req.	Shrunk model $\times 1$	0.60 [0.44, 0.71]	0.7 [0.5, 0.9]	0/36
Plurality	Observed $\times 1$	0.14 [0.09, 0.20]	1.9 [1.5, 2.4]	0/36
Plurality	Observed $\times 1.5$	0.10 [0.07, 0.14]	2.8 [2.4, 3.5]	0/36
Plurality	Observed $\times 2$	0.08 [0.05, 0.12]	3.9 [3.1, 4.9]	0/36
Plurality	Shrunk model $\times 1$	0.25 [0.18, 0.31]	1.5 [1.3, 1.7]	0/36

Table 22: Replay sensitivity to a second noise seed on fixed targets and to three redrawn target sets. Entries are medians across the stated targets; brackets are 95% target-bootstrap intervals. Every run uses 200 draws and the observed-centred SE $\times 1$ model.

Rule	Run	Targets	p_0 [95%]	x_{90} [95%]
OLLB; 200 noise draws per target				
Mean	Fixed grid, noise study seed	36	0.43 [0.42, 0.45]	1.3 [1.2, 1.3]
Mean	Fixed grid, noise seed 1	36	0.42 [0.40, 0.43]	1.3 [1.3, 1.4]
Mean	Random tail, seed 1	36	0.41 [0.40, 0.42]	1.4 [1.3, 1.4]
Mean	Random tail, seed 2	36	0.41 [0.40, 0.42]	1.4 [1.4, 1.4]
Mean	Random tail, study seed	36	0.44 [0.42, 0.45]	1.3 [1.2, 1.3]
Geometric	Fixed grid, noise study seed	36	0.40 [0.38, 0.41]	1.7 [1.7, 1.8]
Geometric	Fixed grid, noise seed 1	36	0.39 [0.38, 0.41]	1.8 [1.8, 1.9]
Geometric	Random tail, seed 1	36	0.38 [0.37, 0.39]	1.8 [1.7, 1.8]
Geometric	Random tail, seed 2	36	0.39 [0.37, 0.40]	1.9 [1.8, 1.9]
Geometric	Random tail, study seed	36	0.40 [0.38, 0.41]	1.8 [1.7, 1.8]
Harmonic	Fixed grid, noise study seed	36	0.34 [0.30, 0.34]	2.5 [2.5, 2.6]
Harmonic	Fixed grid, noise seed 1	36	0.34 [0.33, 0.36]	2.4 [2.4, 2.6]
Harmonic	Random tail, seed 1	36	0.33 [0.31, 0.36]	2.5 [2.4, 2.6]
Harmonic	Random tail, seed 2	36	0.33 [0.32, 0.34]	2.7 [2.6, 2.8]
Harmonic	Random tail, study seed	36	0.33 [0.31, 0.34]	2.6 [2.5, 2.7]
Median	Fixed grid, noise study seed	36	0.26 [0.23, 0.27]	1.5 [1.4, 1.5]
Median	Fixed grid, noise seed 1	36	0.26 [0.24, 0.29]	1.5 [1.4, 1.6]
Median	Random tail, seed 1	36	0.27 [0.25, 0.28]	1.6 [1.5, 1.6]
Median	Random tail, seed 2	36	0.27 [0.24, 0.30]	1.5 [1.4, 1.6]
Median	Random tail, study seed	36	0.26 [0.24, 0.28]	1.5 [1.4, 1.5]
Minimum	Fixed grid, noise study seed	36	0.14 [0.12, 0.17]	5.6 [5.4, 5.7]
Minimum	Fixed grid, noise seed 1	36	0.13 [0.12, 0.15]	5.7 [5.4, 5.8]
Minimum	Random tail, seed 1	36	0.12 [0.11, 0.16]	5.6 [5.4, 5.8]
Minimum	Random tail, seed 2	36	0.14 [0.10, 0.15]	5.7 [5.7, 6.0]
Minimum	Random tail, study seed	36	0.14 [0.12, 0.18]	5.7 [5.4, 5.8]
HELM; 200 noise draws per target				
Mean	Fixed grid, noise study seed	36	0.24 [0.20, 0.28]	1.3 [0.9, 1.3]
Mean	Fixed grid, noise seed 1	36	0.24 [0.23, 0.28]	1.4 [1.1, 1.4]
Mean	Random tail, seed 1	34	0.23 [0.19, 0.28]	1.4 [1.1, 1.5]
Mean	Random tail, seed 2	31	0.24 [0.20, 0.32]	1.3 [1.0, 1.4]
Mean	Random tail, study seed	33	0.23 [0.17, 0.27]	1.3 [1.1, 1.4]
Geometric	Fixed grid, noise study seed	36	0.21 [0.19, 0.25]	1.4 [1.1, 1.5]
Geometric	Fixed grid, noise seed 1	36	0.23 [0.20, 0.24]	1.5 [1.3, 1.6]
Geometric	Random tail, seed 1	34	0.21 [0.17, 0.24]	1.5 [1.3, 1.7]
Geometric	Random tail, seed 2	31	0.23 [0.21, 0.28]	1.4 [1.2, 1.6]
Geometric	Random tail, study seed	33	0.21 [0.17, 0.23]	1.4 [1.2, 1.5]
Harmonic	Fixed grid, noise study seed	36	0.24 [0.20, 0.26]	1.5 [1.3, 1.6]
Harmonic	Fixed grid, noise seed 1	36	0.23 [0.20, 0.27]	1.6 [1.4, 1.8]
Harmonic	Random tail, seed 1	34	0.22 [0.20, 0.27]	1.6 [1.4, 1.8]
Harmonic	Random tail, seed 2	31	0.25 [0.23, 0.28]	1.6 [1.4, 1.7]
Harmonic	Random tail, study seed	33	0.23 [0.20, 0.24]	1.5 [1.3, 1.7]
Median	Fixed grid, noise study seed	36	0.22 [0.18, 0.28]	2.0 [1.8, 2.2]
Median	Fixed grid, noise seed 1	36	0.24 [0.20, 0.32]	2.0 [1.8, 2.1]
Median	Random tail, seed 1	34	0.27 [0.23, 0.35]	1.8 [1.6, 2.0]
Median	Random tail, seed 2	31	0.20 [0.15, 0.30]	2.1 [1.9, 2.4]
Median	Random tail, study seed	33	0.23 [0.18, 0.29]	1.9 [1.7, 2.3]
Minimum	Fixed grid, noise study seed	36	0.24 [0.19, 0.28]	9.0 [7.6, 9.6]
Minimum	Fixed grid, noise seed 1	36	0.23 [0.18, 0.26]	8.8 [7.8, 9.3]
Minimum	Random tail, seed 1	34	0.20 [0.14, 0.28]	9.1 [8.5, 9.6]
Minimum	Random tail, seed 2	31	0.21 [0.18, 0.26]	9.7 [7.9, 10.1]
Minimum	Random tail, study seed	33	0.18 [0.15, 0.26]	9.3 [8.2, 9.8]
Borda	Fixed grid, noise study seed	36	0.80 [0.65, 0.92]	0.3 [0, 0.7]
Borda	Fixed grid, noise seed 1	36	0.79 [0.66, 0.90]	0.4 [0.016, 0.6]

Rule	Run	Targets	p_0 [95%]	x_{90} [95%]
Borda	Random tail, seed 1	34	0.87 [0.68, 0.93]	0.2 [0, 0.6]
Borda	Random tail, seed 2	31	0.85 [0.64, 0.93]	0.2 [0, 0.7]
Borda	Random tail, study seed	33	0.88 [0.71, 0.93]	0.1 [0, 0.5]
Condorcet req.	Fixed grid, noise study seed	36	0.29 [0.20, 0.35]	1.2 [1.0, 1.4]
Condorcet req.	Fixed grid, noise seed 1	36	0.28 [0.21, 0.35]	1.2 [1.1, 1.4]
Condorcet req.	Random tail, seed 1	34	0.21 [0.14, 0.28]	1.4 [1.1, 1.5]
Condorcet req.	Random tail, seed 2	31	0.23 [0.19, 0.31]	1.3 [1.1, 1.5]
Condorcet req.	Random tail, study seed	33	0.22 [0.17, 0.30]	1.3 [1.1, 1.4]
Plurality	Fixed grid, noise study seed	36	0.14 [0.09, 0.20]	1.9 [1.5, 2.4]
Plurality	Fixed grid, noise seed 1	36	0.11 [0.07, 0.15]	1.9 [1.5, 2.3]
Plurality	Random tail, seed 1	34	0.12 [0.07, 0.17]	1.9 [1.4, 2.3]
Plurality	Random tail, seed 2	31	0.16 [0.08, 0.20]	1.5 [1.3, 1.9]
Plurality	Random tail, study seed	33	0.12 [0.08, 0.20]	1.8 [1.4, 2.4]

Table 23: Reoptimized gain10 cost sensitivity on the same 10 HELM or 20 OLLB targets (mean ranks 2–11 or 2–21). Each cell summarizes per-target draw summaries; brackets are 95% target-bootstrap intervals, not draw quantiles or fixed-strategy replay success.

Rule	Draws	Median L_1 [95%]	Feas. draws (%) [95%]	Cost ratio [95%]
HELM, evaluation noise, delivered (seed 0)				
Mean	100	174.0 [48.3, 235.3]	100.0 [100.0, 100.0]	1.06 [1.04, 1.12]
Geometric	100	170.0 [46.3, 227.3]	100.0 [100.0, 100.0]	1.08 [1.06, 1.21]
Median	100	6.5 [2.2, 18.3]	100.0 [100.0, 100.0]	1.23 [1.05, 8.08]
Minimum	100	7.0 [4.2, 20.0]	72.5 [12.0, 84.0]	1.60 [0.00, 2.55]
HELM, evaluation noise, seed 20261108				
Mean	200	171.9 [52.4, 232.7]	100.0 [100.0, 100.0]	1.06 [1.02, 1.14]
Geometric	200	167.1 [49.7, 222.8]	100.0 [100.0, 100.0]	1.08 [1.04, 1.30]
Median	200	6.0 [2.6, 18.1]	100.0 [100.0, 100.0]	1.27 [1.12, 9.12]
Minimum	200	8.3 [3.5, 18.0]	77.2 [14.0, 88.5]	1.40 [0.00, 2.77]
HELM, task bootstrap, delivered (seed 0)				
Mean	100	174.8 [50.2, 230.9]	100.0 [100.0, 100.0]	1.07 [1.04, 1.20]
Geometric	100	170.2 [45.8, 245.2]	100.0 [100.0, 100.0]	1.08 [1.05, 1.20]
Median	100	6.5 [2.4, 19.9]	100.0 [100.0, 100.0]	1.29 [1.14, 6.47]
Minimum	100	4.0 [1.5, 10.6]	82.5 [37.0, 99.0]	1.00 [0.00, 1.60]
HELM, task bootstrap, seed 20261108				
Mean	200	168.2 [47.6, 226.9]	100.0 [100.0, 100.0]	1.04 [1.02, 1.22]
Geometric	200	165.6 [46.4, 218.4]	100.0 [100.0, 100.0]	1.06 [1.03, 1.21]
Median	200	6.7 [1.4, 19.5]	100.0 [100.0, 100.0]	1.29 [1.12, 3.96]
Minimum	200	4.1 [1.6, 10.2]	84.0 [38.0, 98.5]	1.00 [0.00, 1.60]
OLLB, evaluation noise, delivered (seed 0)				
Mean	100	30.0 [27.3, 32.4]	100.0 [100.0, 100.0]	1.03 [1.02, 1.04]
Geometric	100	25.5 [19.9, 28.7]	100.0 [99.0, 100.0]	1.05 [1.04, 1.05]
Median	100	4.4 [2.4, 10.1]	100.0 [100.0, 100.0]	1.22 [1.08, 1.56]
Minimum	100	13.2 [9.3, 15.2]	54.5 [30.5, 85.0]	1.22 [1.00, 1.56]
OLLB, evaluation noise, seed 20261108				
Mean	200	30.1 [27.4, 32.0]	100.0 [100.0, 100.0]	1.03 [1.03, 1.05]
Geometric	200	26.4 [20.4, 29.9]	100.0 [99.2, 100.0]	1.09 [1.09, 1.10]
Median	200	4.6 [2.3, 10.1]	100.0 [100.0, 100.0]	1.25 [1.08, 1.58]
Minimum	200	12.7 [9.7, 14.8]	58.5 [29.2, 87.8]	1.19 [0.99, 1.51]
OLLB, task bootstrap, delivered (seed 0)				
Mean	200	28.6 [25.6, 31.1]	83.5 [81.0, 86.5]	1.02 [0.96, 1.10]
Geometric	200	23.1 [19.3, 25.4]	79.0 [72.2, 84.0]	1.01 [0.85, 1.12]
Median	200	8.0 [6.8, 9.6]	62.3 [60.5, 62.5]	2.61 [1.00, 6.56]
Minimum	200	7.4 [5.2, 14.5]	51.5 [47.8, 77.5]	1.40 [1.18, 1.48]

Rule	Draws	Median L_1 [95%]	Feas. draws (%) [95%]	Cost ratio [95%]
OLLB, task bootstrap, seed 20261108				
Mean	200	29.6 [27.4, 32.4]	83.2 [81.0, 86.0]	1.03 [1.00, 1.14]
Geometric	200	24.1 [20.7, 26.4]	74.5 [68.7, 80.5]	1.10 [0.87, 1.12]
Median	200	7.0 [5.0, 10.8]	63.0 [58.7, 63.5]	2.19 [1.09, 3.49]
Minimum	200	7.6 [5.6, 14.5]	48.5 [45.8, 70.8]	1.37 [1.16, 1.45]

Table 24: OLLB ceiling-cost medians and mean/minimum feasibility: identical point estimates with individual-target and base-family 95% percentile-bootstrap intervals side by side. Feasibility entries are percentages.

Rule	Statistic	Estimate	Individual 95%	Family 95%
Mean	K, ceil	2.0	[2.0, 2.0]	[2.0, 2.0]
Mean	L_1, ceil	180.5	[177.9, 182.3]	[176.0, 184.4]
Mean	Feas. gain10 (%)	2.5	[2.1, 3.0]	[1.8, 3.3]
Mean	Feas. gain20 (%)	18.5	[17.4, 19.6]	[16.6, 20.5]
Geometric	K, ceil	3.0	[3.0, 3.0]	[3.0, 3.0]
Geometric	L_1, ceil	153.2	[150.7, 154.7]	[149.0, 156.7]
Harmonic	K, ceil	3.0	[3.0, 3.0]	[3.0, 3.0]
Harmonic	L_1, ceil	124.8	[123.7, 126.4]	[121.8, 128.2]
Trimmed	K, ceil	3.0	[3.0, 3.0]	[3.0, 3.0]
Trimmed	L_1, ceil	141.1	[139.9, 142.4]	[138.5, 144.2]
Median	K, ceil	3.0	[3.0, 3.0]	[3.0, 3.0]
Median	L_1, ceil	105.9	[104.8, 107.4]	[103.6, 109.1]
Minimum	K, ceil	3.0	[3.0, 3.0]	[3.0, 3.0]
Minimum	L_1, ceil	43.2	[42.5, 44.1]	[41.5, 45.3]
Minimum	Feas. gain10 (%)	9.9	[9.1, 10.9]	[8.7, 11.3]
Minimum	Feas. gain20 (%)	63.9	[62.5, 65.3]	[60.3, 67.0]

Table 25: OLLB sensitivity to family-size exclusions. Ceiling costs and gain20 feasible shares carry 95% base-family-cluster intervals; ratio is the ratio of median L_1 to the mean rule on the same subset. The last block has only seven families, so its percentile intervals are crude.

Rule	Targets / families	L_1 [95%]	Ratio [95%]	Feas. gain20 (%)
All families				
Mean	4495 / 3849	180.5 [176.0, 184.4]	1 (reference)	18.5% [16.6, 20.5]
Geometric	4495 / 3849	153.2 [149.0, 156.7]	0.849 [0.846, 0.851]	21.3% [19.2, 23.6]
Harmonic	4495 / 3849	124.8 [121.8, 128.2]	0.691 [0.689, 0.698]	25.7% [23.4, 28.2]
Trimmed	4495 / 3849	141.1 [138.5, 144.2]	0.782 [0.778, 0.790]	13.5% [11.9, 15.5]
Median	4495 / 3849	105.9 [103.6, 109.1]	0.587 [0.585, 0.592]	14.3% [12.6, 16.3]
Minimum	4495 / 3849	43.2 [41.5, 45.3]	0.239 [0.235, 0.246]	63.9% [60.3, 67.0]
Families of size ≤ 10				
Mean	4033 / 3832	178.2 [176.1, 181.4]	1 (reference)	18.9% [17.5, 20.3]
Geometric	4033 / 3832	151.2 [149.0, 153.7]	0.848 [0.846, 0.851]	21.9% [20.5, 23.3]
Harmonic	4033 / 3832	123.7 [121.6, 125.2]	0.694 [0.689, 0.698]	26.6% [25.1, 28.1]
Trimmed	4033 / 3832	139.9 [138.2, 141.4]	0.785 [0.779, 0.789]	13.7% [12.5, 14.9]
Median	4033 / 3832	105.0 [103.3, 106.5]	0.589 [0.585, 0.592]	14.4% [13.3, 15.6]
Minimum	4033 / 3832	42.5 [41.5, 43.2]	0.238 [0.235, 0.241]	65.4% [63.8, 67.0]
Families of size ≤ 20				
Mean	4193 / 3842	177.8 [175.7, 180.9]	1 (reference)	19.0% [17.5, 20.6]
Geometric	4193 / 3842	150.7 [148.6, 153.6]	0.847 [0.845, 0.850]	21.9% [20.3, 23.5]
Harmonic	4193 / 3842	123.4 [121.1, 125.0]	0.694 [0.688, 0.697]	26.5% [24.8, 28.2]
Trimmed	4193 / 3842	139.7 [138.1, 141.2]	0.785 [0.779, 0.790]	14.0% [12.6, 15.4]
Median	4193 / 3842	104.8 [103.0, 106.3]	0.589 [0.585, 0.592]	14.6% [13.3, 16.1]
Minimum	4193 / 3842	42.4 [41.3, 43.4]	0.238 [0.234, 0.242]	65.9% [64.3, 67.4]

Rule	Targets / families	L_1 [95%]	Ratio [95%]	Feas. gain20 (%)
Families of size ≤ 50				
Mean	4325 / 3847	178.0 [175.4, 181.5]	1 (reference)	19.2% [17.5, 21.0]
Geometric	4325 / 3847	150.8 [148.4, 153.8]	0.847 [0.845, 0.850]	22.2% [20.3, 24.1]
Harmonic	4325 / 3847	123.7 [121.1, 125.5]	0.695 [0.689, 0.699]	26.8% [24.7, 28.8]
Trimmed	4325 / 3847	139.9 [138.0, 141.8]	0.786 [0.779, 0.791]	14.1% [12.5, 15.9]
Median	4325 / 3847	104.8 [102.9, 106.4]	0.589 [0.585, 0.592]	14.8% [13.2, 16.6]
Minimum	4325 / 3847	42.5 [41.2, 43.7]	0.239 [0.234, 0.243]	65.4% [63.6, 67.2]
Only families of size > 20				
Mean	302 / 7	228.8 [145.3, 265.7]	1 (reference)	11.3% [0.0, 36.5]
Geometric	302 / 7	200.8 [121.8, 237.7]	0.878 [0.817, 0.894]	13.9% [0.0, 44.4]
Harmonic	302 / 7	170.5 [93.8, 207.4]	0.745 [0.627, 0.781]	15.2% [0.0, 48.2]
Trimmed	302 / 7	185.5 [107.5, 218.8]	0.811 [0.771, 0.825]	7.9% [0.0, 32.4]
Median	302 / 7	148.7 [78.8, 180.0]	0.650 [0.549, 0.677]	9.3% [0.0, 34.4]
Minimum	302 / 7	64.6 [34.8, 94.0]	0.282 [0.229, 0.354]	36.8% [10.9, 73.1]

Table 26: Ceiling-cost medians by OLLB parameter-size bin and mean-rank band, with 95% base-family-cluster percentile intervals. Targets / families gives each stratum denominator; all ceiling targets are feasible.

Rule	Targets / families	K [95%]	L_1 [95%]
Parameters: $\leq$*3B*			
Mean	779 / 572	3.0 [3.0, 3.0]	265.4 [259.5, 266.6]
Geometric	779 / 572	4.0 [4.0, 4.0]	237.4 [231.5, 238.6]
Harmonic	779 / 572	5.0 [5.0, 5.0]	207.1 [201.2, 208.3]
Trimmed	779 / 572	3.0 [3.0, 3.0]	218.5 [212.3, 219.7]
Median	779 / 572	3.0 [3.0, 3.0]	179.6 [173.9, 180.7]
Minimum	779 / 572	5.0 [5.0, 5.0]	94.4 [93.4, 95.5]
Parameters: 3–7B			
Mean	464 / 404	3.0 [3.0, 3.0]	204.1 [195.2, 211.9]
Geometric	464 / 404	3.0 [3.0, 3.0]	176.0 [167.2, 183.9]
Harmonic	464 / 404	4.0 [3.0, 4.0]	147.7 [139.8, 154.8]
Trimmed	464 / 404	3.0 [3.0, 3.0]	160.9 [154.0, 168.5]
Median	464 / 404	3.0 [3.0, 3.0]	124.8 [119.5, 132.0]
Minimum	464 / 404	5.0 [4.0, 5.0]	54.5 [50.3, 57.5]
Parameters: 7–13B			
Mean	2275 / 2051	2.0 [2.0, 2.0]	175.2 [173.1, 176.9]
Geometric	2275 / 2051	3.0 [3.0, 3.0]	147.9 [146.2, 149.8]
Harmonic	2275 / 2051	3.0 [3.0, 3.0]	120.2 [117.9, 122.6]
Trimmed	2275 / 2051	3.0 [3.0, 3.0]	137.6 [135.7, 139.1]
Median	2275 / 2051	3.0 [3.0, 3.0]	102.2 [100.8, 103.7]
Minimum	2275 / 2051	3.0 [3.0, 3.0]	41.1 [40.4, 42.1]
Parameters: 13–35B			
Mean	770 / 683	2.0 [2.0, 2.0]	91.7 [87.4, 96.1]
Geometric	770 / 683	2.0 [2.0, 2.0]	66.2 [62.7, 71.7]
Harmonic	770 / 683	2.0 [2.0, 2.0]	44.8 [41.2, 50.6]
Trimmed	770 / 683	2.0 [2.0, 2.0]	67.2 [62.2, 73.2]
Median	770 / 683	3.0 [3.0, 3.0]	40.7 [38.3, 46.8]
Minimum	770 / 683	2.0 [2.0, 2.0]	20.3 [19.1, 21.7]
Parameters: $>$ *35B*			
Mean	197 / 153	2.0 [2.0, 2.0]	96.6 [90.2, 110.6]
Geometric	197 / 153	2.0 [2.0, 2.0]	72.4 [66.5, 84.8]
Harmonic	197 / 153	2.0 [2.0, 2.0]	52.9 [46.5, 62.8]
Trimmed	197 / 153	2.0 [2.0, 2.0]	74.3 [69.1, 82.3]
Median	197 / 153	2.0 [2.0, 3.0]	48.4 [41.4, 55.8]
Minimum	197 / 153	2.0 [2.0, 3.0]	18.8 [15.3, 22.0]

Rule	Targets / families	K [95%]	L_1 [95%]
Parameters: unknown			
Mean	10 / 10	3.0 [2.0, 3.0]	205.0 [140.8, 228.5]
Geometric	10 / 10	3.0 [2.0, 4.0]	176.9 [112.8, 200.5]
Harmonic	10 / 10	4.0 [2.0, 4.0]	146.7 [83.8, 170.2]
Trimmed	10 / 10	3.0 [2.0, 3.0]	165.1 [103.0, 185.1]
Median	10 / 10	3.0 [3.0, 3.0]	133.5 [70.3, 149.6]
Minimum	10 / 10	4.0 [2.0, 6.0]	48.5 [29.5, 54.9]
Mean ranks: 2–20			
Mean	19 / 14	1.0 [1.0, 1.0]	29.1 [25.4, 31.7]
Geometric	19 / 14	1.0 [1.0, 1.0]	21.7 [16.1, 23.8]
Harmonic	19 / 14	1.0 [1.0, 1.0]	17.7 [7.2, 21.8]
Trimmed	19 / 14	1.0 [1.0, 1.0]	24.3 [20.9, 25.6]
Median	19 / 14	1.0 [1.0, 1.0]	3.3 [1.0, 9.6]
Minimum	19 / 14	2.0 [1.0, 2.0]	10.8 [3.7, 15.7]
Mean ranks: 21–200			
Mean	180 / 156	1.0 [1.0, 1.0]	59.3 [57.2, 61.5]
Geometric	180 / 156	1.0 [1.0, 1.0]	41.1 [40.1, 42.3]
Harmonic	180 / 156	2.0 [1.0, 2.0]	28.2 [26.6, 29.3]
Trimmed	180 / 156	1.0 [1.0, 1.0]	47.0 [44.9, 49.3]
Median	180 / 156	2.0 [2.0, 2.0]	20.9 [18.5, 21.4]
Minimum	180 / 156	2.0 [2.0, 2.0]	14.7 [11.9, 16.3]
Mean ranks: 201+			
Mean	4296 / 3690	2.0 [2.0, 3.0]	183.1 [179.6, 187.9]
Geometric	4296 / 3690	3.0 [3.0, 3.0]	155.8 [152.2, 160.1]
Harmonic	4296 / 3690	3.0 [3.0, 3.0]	127.2 [124.3, 131.2]
Trimmed	4296 / 3690	3.0 [3.0, 3.0]	143.0 [140.5, 146.5]
Median	4296 / 3690	3.0 [3.0, 3.0]	108.1 [105.6, 111.1]
Minimum	4296 / 3690	3.0 [3.0, 3.0]	44.6 [42.9, 46.6]

Table 27: Gain-cap feasible percentages by OLLB parameter-size bin and mean-rank band, with 95% base-family-cluster percentile intervals. Targets / families gives each stratum denominator; all ceiling targets are feasible.

Rule	Targets / families	Feas. gain10 (%)	Feas. gain20 (%)
Parameters: $\leq$3B			
Mean	779 / 572	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Geometric	779 / 572	0.0 [0.0, 0.0]	0.1 [0.0, 0.5]
Harmonic	779 / 572	0.0 [0.0, 0.0]	0.3 [0.0, 0.7]
Trimmed	779 / 572	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Median	779 / 572	0.0 [0.0, 0.0]	0.1 [0.0, 0.5]
Minimum	779 / 572	0.0 [0.0, 0.0]	14.5 [10.3, 19.8]
Parameters: 3–7B			
Mean	464 / 404	0.0 [0.0, 0.0]	0.2 [0.0, 0.7]
Geometric	464 / 404	0.0 [0.0, 0.0]	0.4 [0.0, 1.1]
Harmonic	464 / 404	0.0 [0.0, 0.0]	4.1 [2.3, 6.1]
Trimmed	464 / 404	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Median	464 / 404	0.0 [0.0, 0.0]	0.2 [0.0, 0.7]
Minimum	464 / 404	0.0 [0.0, 0.0]	44.8 [38.7, 51.1]
Parameters: 7–13B			
Mean	2275 / 2051	0.0 [0.0, 0.0]	10.0 [8.5, 11.6]
Geometric	2275 / 2051	0.0 [0.0, 0.0]	13.5 [11.7, 15.4]
Harmonic	2275 / 2051	0.0 [0.0, 0.1]	18.7 [16.6, 21.0]
Trimmed	2275 / 2051	0.0 [0.0, 0.0]	2.4 [1.8, 3.1]
Median	2275 / 2051	0.1 [0.0, 0.3]	5.1 [4.2, 6.2]
Minimum	2275 / 2051	0.6 [0.3, 0.9]	75.1 [73.3, 77.0]

Rule	Targets / families	Feas. gain10 (%)	Feas. gain20 (%)
Parameters: 13–35B			
Mean	770 / 683	7.7 [5.3, 10.3]	61.4 [57.5, 65.2]
Geometric	770 / 683	11.6 [9.0, 14.3]	67.0 [62.9, 71.0]
Harmonic	770 / 683	26.8 [23.1, 30.3]	73.6 [69.9, 77.1]
Trimmed	770 / 683	3.9 [1.9, 6.5]	57.9 [53.4, 62.0]
Median	770 / 683	22.2 [18.9, 25.4]	56.2 [51.6, 60.6]
Minimum	770 / 683	44.4 [40.7, 48.2]	86.6 [84.0, 89.0]
Parameters: $>$ 35B			
Mean	197 / 153	26.9 [18.1, 35.3]	65.0 [56.0, 73.0]
Geometric	197 / 153	26.4 [17.5, 35.1]	67.5 [58.9, 75.4]
Harmonic	197 / 153	28.4 [18.8, 38.0]	72.1 [63.9, 79.3]
Trimmed	197 / 153	12.2 [5.8, 19.6]	54.3 [43.8, 63.8]
Median	197 / 153	14.7 [8.5, 22.0]	44.7 [34.1, 54.7]
Minimum	197 / 153	45.7 [37.1, 54.0]	86.3 [80.9, 90.8]
Parameters: unknown			
Mean	10 / 10	10.0 [0.0, 30.0]	10.0 [0.0, 30.0]
Geometric	10 / 10	10.0 [0.0, 30.0]	10.0 [0.0, 30.0]
Harmonic	10 / 10	10.0 [0.0, 30.0]	20.0 [0.0, 50.0]
Trimmed	10 / 10	10.0 [0.0, 30.0]	10.0 [0.0, 30.0]
Median	10 / 10	10.0 [0.0, 30.0]	10.0 [0.0, 30.0]
Minimum	10 / 10	10.0 [0.0, 30.0]	60.0 [30.0, 90.0]
Mean ranks: 2–20			
Mean	19 / 14	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
Geometric	19 / 14	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
Harmonic	19 / 14	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
Trimmed	19 / 14	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
Median	19 / 14	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
Minimum	19 / 14	78.9 [57.1, 100.0]	100.0 [100.0, 100.0]
Mean ranks: 21–200			
Mean	180 / 156	52.2 [43.8, 60.6]	100.0 [100.0, 100.0]
Geometric	180 / 156	56.7 [47.7, 65.0]	100.0 [100.0, 100.0]
Harmonic	180 / 156	56.1 [47.2, 65.0]	100.0 [100.0, 100.0]
Trimmed	180 / 156	20.0 [12.4, 29.3]	100.0 [100.0, 100.0]
Median	180 / 156	64.4 [56.1, 72.6]	100.0 [100.0, 100.0]
Minimum	180 / 156	65.6 [56.4, 74.3]	99.4 [98.3, 100.0]
Mean ranks: 201+			
Mean	4296 / 3690	0.0 [0.0, 0.0]	14.7 [13.1, 16.3]
Geometric	4296 / 3690	0.5 [0.3, 0.7]	17.7 [15.8, 19.6]
Harmonic	4296 / 3690	3.4 [2.8, 4.0]	22.3 [20.1, 24.5]
Trimmed	4296 / 3690	0.0 [0.0, 0.0]	9.5 [8.3, 11.0]
Median	4296 / 3690	1.6 [1.2, 2.1]	10.3 [9.0, 11.8]
Minimum	4296 / 3690	7.3 [6.3, 8.4]	62.3 [58.1, 65.4]

Table 28: The ten largest OLLB base-model families and the board-wide family-size distribution. Counts describe the observed board, with no sampling interval. Targets exclude the arithmetic-mean leader.

Family identifier	Models on board	Targets
`Qwen/Qwen2.5-0.5B`	112	112
`meta-llama/Meta-Llama-3.1-8B`	58	58
`Qwen/Qwen2.5-14B`	29	29
`meta-llama/Llama-3.2-3B-Instruct`	29	29
`Qwen/Qwen2.5-7B`	26	26
`mistralai/Mistral-7B-v0.1`	26	26
`HuggingFaceTB/SmolLM2-135M`	22	22
`meta-llama/Meta-Llama-3-8B`	20	20
`meta-llama/Meta-Llama-3-8B-Instruct`	20	20
`Qwen/Qwen2.5-3B`	19	19
All families (board)		3850
Families with targets		3849
Singleton families (board)		3744
Families with 2–5 models		79
Families with 6–20 models		20
Families with $>$ 20 models		7

Table 29: Ratios of ceiling-cap target medians to the mean rule in each score currency. Brackets are 95% paired-target-bootstrap intervals. L_1 and L_∞ are independently minimized; normalization is by each rule-specific score gap or IQR, as defined in the caption notes.

Rule	Targets	L_1 ratio [95% interval]	L_∞ ratio [95% interval]
HELM, 78; Raw task points			
Geometric	78	0.960 [0.953, 0.967]	1.075 [1.042, 1.094]
Harmonic	78	0.924 [0.909, 0.937]	1.162 [1.105, 1.274]
Median	78	0.161 [0.125, 0.195]	0.743 [0.691, 0.848]
Minimum	78	0.141 [0.099, 0.182]	1.689 [1.406, 2.042]
Borda	78	0.766 [0.712, 0.809]	1.146 [1.046, 1.255]
Condorcet req.	78	0.306 [0.271, 0.331]	0.777 [0.707, 0.839]
Plurality	78	0.068 [0.058, 0.076]	0.471 [0.426, 0.546]
HELM, 78; Per own-rule gap			
Geometric	78	0.877 [0.871, 0.887]	0.975 [0.970, 0.983]
Harmonic	78	0.764 [0.753, 0.777]	0.947 [0.938, 0.957]
Median	78	0.188 [0.169, 0.231]	0.938 [0.930, 0.950]
Minimum	78	0.088 [0.069, 0.109]	0.938 [0.930, 0.950]
HELM, 78; Per own-rule IQR			
Geometric	78	0.860 [0.853, 0.866]	0.963 [0.933, 0.979]
Harmonic	78	0.740 [0.729, 0.751]	0.931 [0.885, 1.021]
Median	78	0.167 [0.129, 0.202]	0.768 [0.714, 0.877]
Minimum	78	0.096 [0.068, 0.124]	1.154 [0.961, 1.395]
OLLB census, 4495; Raw task points			
Geometric	4495	0.849 [0.846, 0.851]	0.922 [0.918, 0.930]
Harmonic	4495	0.691 [0.689, 0.696]	0.827 [0.822, 0.834]
Median	4495	0.587 [0.584, 0.591]	1.202 [1.194, 1.212]
Minimum	4495	0.239 [0.237, 0.243]	0.595 [0.590, 0.602]
OLLB census, 4495; Per own-rule gap			
Geometric	4495	0.842 [0.838, 0.846]	0.913 [0.912, 0.915]
Harmonic	4495	0.710 [0.706, 0.714]	0.849 [0.846, 0.851]
Median	4495	0.496 [0.493, 0.499]	1.000 [1.000, 1.000]
Minimum	4495	0.414 [0.410, 0.418]	1.000 [1.000, 1.000]
OLLB census, 4495; Per own-rule IQR			
Geometric	4495	0.951 [0.948, 0.953]	1.034 [1.028, 1.042]
Harmonic	4495	0.852 [0.849, 0.858]	1.019 [1.013, 1.027]
Median	4495	0.568 [0.566, 0.572]	1.164 [1.156, 1.174]
Minimum	4495	0.536 [0.531, 0.544]	1.332 [1.323, 1.349]

Rule	Targets	L_1 ratio [95% interval]	L_∞ ratio [95% interval]
OLLB sample, 136; Raw task points			
Geometric	136	0.692 [0.673, 0.744]	1.013 [0.956, 1.089]
Harmonic	136	0.493 [0.463, 0.537]	0.990 [0.929, 1.108]
Trimmed	136	0.824 [0.763, 0.874]	1.236 [1.144, 1.311]
Median	136	0.364 [0.341, 0.406]	1.028 [0.917, 1.136]
Minimum	136	0.293 [0.250, 0.332]	0.973 [0.867, 1.120]
Borda	136	0.436 [0.402, 0.500]	0.813 [0.720, 0.912]
Condorcet req.	136	0.627 [0.606, 0.685]	1.344 [1.259, 1.446]
Plurality	136	0.444 [0.417, 0.487]	1.677 [1.570, 1.813]
OLLB sample, 136; Per own-rule gap			
Geometric	136	0.616 [0.595, 0.643]	0.879 [0.868, 0.886]
Harmonic	136	0.397 [0.381, 0.415]	0.788 [0.777, 0.800]
Trimmed	136	0.667 [0.667, 0.667]	1.000 [1.000, 1.000]
Median	136	0.367 [0.336, 0.384]	1.000 [1.000, 1.000]
Minimum	136	0.296 [0.285, 0.306]	1.000 [1.000, 1.000]
OLLB sample, 136; Per own-rule IQR			
Geometric	136	0.775 [0.754, 0.833]	1.135 [1.071, 1.220]
Harmonic	136	0.607 [0.571, 0.661]	1.220 [1.144, 1.365]
Trimmed	136	0.879 [0.814, 0.932]	1.318 [1.220, 1.399]
Median	136	0.352 [0.330, 0.393]	0.995 [0.888, 1.100]
Minimum	136	0.657 [0.560, 0.743]	2.181 [1.944, 2.509]

Table 30: Ratios of median ceiling L_1 to the mean within bins of rank under each rule. Brackets are 95% target-bootstrap ratio intervals. The target cohorts in a bin can differ between rules; bins without a supplied mean comparison are omitted.

Rule	Own-rule rank bin	Targets	L_1 ratio [95% interval]
HELM 78 targets			
Geometric	3–5	3	0.916 [0.086, 6.181]
Geometric	6–10	5	0.927 [0.625, 1.354]
Geometric	11–20	10	0.937 [0.737, 1.173]
Geometric	21–80	59	0.968 [0.724, 1.278]
Harmonic	3–5	3	0.908 [0.060, 5.598]
Harmonic	6–10	5	0.875 [0.577, 1.362]
Harmonic	11–20	10	0.883 [0.676, 1.126]
Harmonic	21–80	59	0.937 [0.696, 1.260]
Median	3–5	3	0.009 [0.002, 0.068]
Median	6–10	5	0.076 [0.018, 0.104]
Median	11–20	10	0.096 [0.054, 0.115]
Median	21–80	59	0.194 [0.128, 0.273]
Minimum	6–10	5	0.043 [0.017, 0.088]
Minimum	11–20	10	0.103 [0.059, 0.140]
Minimum	21–80	59	0.175 [0.122, 0.261]
Borda	3–5	3	0.505 [0.036, 2.447]
Borda	6–10	5	0.320 [0.208, 0.774]
Borda	11–20	10	0.609 [0.438, 0.779]
Borda	21–80	59	0.812 [0.589, 1.097]
Condorcet req.	3–5	5	0.395 [0.021, 0.917]
Condorcet req.	6–10	3	0.313 [0.132, 0.370]
Condorcet req.	11–20	11	0.278 [0.193, 0.330]
Condorcet req.	21–80	58	0.314 [0.223, 0.402]
Plurality	6–10	5	0.043 [0.023, 0.086]
Plurality	11–20	69	0.147 [0.113, 0.194]

Rule	Own-rule rank bin	Targets	L_1 ratio [95% interval]
OLLB 136-sample			
Geometric	2–5	5	0.563 [0.154, 3.257]
Geometric	6–10	4	0.721 [0.654, 0.857]
Geometric	11–21	9	0.683 [0.670, 0.778]
Geometric	22–100	34	0.639 [0.565, 0.729]
Geometric	101–1000	58	0.648 [0.552, 0.760]
Geometric	1001–5000	26	0.854 [0.746, 1.009]
Harmonic	2–5	5	0.300 [0.081, 1.461]
Harmonic	11–21	7	0.439 [0.421, 0.472]
Harmonic	22–100	22	0.398 [0.366, 0.485]
Harmonic	101–1000	71	0.415 [0.349, 0.466]
Harmonic	1001–5000	29	0.671 [0.576, 0.801]
Trimmed	2–5	4	1.012 [0.276, 3.139]
Trimmed	6–10	5	0.772 [0.742, 0.869]
Trimmed	11–21	11	0.771 [0.763, 0.838]
Trimmed	22–100	38	0.786 [0.693, 0.910]
Trimmed	101–1000	50	0.733 [0.628, 0.836]
Trimmed	1001–5000	28	0.790 [0.673, 0.908]
Median	2–5	4	0.081 [0.002, 0.149]
Median	6–10	4	0.067 [0.037, 0.092]
Median	11–21	11	0.106 [0.091, 0.127]
Median	22–100	29	0.296 [0.207, 0.330]
Median	101–1000	60	0.343 [0.291, 0.397]
Median	1001–5000	27	0.596 [0.495, 0.708]
Minimum	2–5	3	0.136 [0.037, 0.186]
Minimum	11–21	3	0.098 [0.087, 0.135]
Minimum	22–100	11	0.131 [0.097, 0.158]
Minimum	101–1000	84	0.207 [0.168, 0.250]
Minimum	1001–5000	34	0.229 [0.174, 0.267]
Borda	2–5	3	0.254 [0.047, 0.931]
Borda	6–10	5	0.182 [0.105, 0.264]
Borda	11–21	9	0.271 [0.179, 0.373]
Borda	22–100	42	0.295 [0.262, 0.399]
Borda	101–1000	46	0.481 [0.396, 0.614]
Borda	1001–5000	30	0.775 [0.649, 0.905]
Plurality	6–10	134	0.984 [0.908, 1.162]
OLLB census 4495			
Geometric	3–5	4	1.249 [0.139, 3.150]
Geometric	6–10	4	0.721 [0.654, 0.878]
Geometric	11–20	10	0.738 [0.677, 0.867]
Geometric	21–50	30	0.738 [0.606, 0.820]
Geometric	51–200	150	0.708 [0.685, 0.734]
Geometric	201–1000	800	0.785 [0.749, 0.814]
Geometric	1001–5000	3496	0.859 [0.841, 0.877]
Harmonic	3–5	4	0.580 [0.077, 1.413]
Harmonic	6–10	4	0.413 [0.396, 0.472]
Harmonic	11–20	10	0.478 [0.433, 0.550]
Harmonic	21–50	30	0.509 [0.419, 0.556]
Harmonic	51–200	150	0.451 [0.440, 0.469]
Harmonic	201–1000	800	0.594 [0.553, 0.622]
Harmonic	1001–5000	3496	0.709 [0.694, 0.724]
Median	3–5	3	0.089 [0.015, 0.151]
Median	6–10	5	0.077 [0.037, 0.088]
Median	11–20	10	0.104 [0.092, 0.124]
Median	21–50	29	0.236 [0.193, 0.279]
Median	51–200	150	0.333 [0.320, 0.348]
Median	201–1000	800	0.490 [0.469, 0.519]
Median	1001–5000	3496	0.600 [0.590, 0.614]
Minimum	3–5	3	0.112 [0.037, 0.394]
Minimum	6–10	6	0.082 [0.062, 0.115]
Minimum	11–20	8	0.098 [0.094, 0.128]

Rule	Own-rule rank bin	Targets	L_1 ratio [95% interval]
Minimum	21–50	30	0.147 [0.127, 0.165]
Minimum	51–200	153	0.137 [0.124, 0.143]
Minimum	201–1000	805	0.218 [0.211, 0.223]
Minimum	1001–5000	3488	0.256 [0.249, 0.263]

Table 31: OLLB minimum/mean comparison after separating targets with zero task scores and removing zero-score models. Ceiling ratios carry 95% target-bootstrap intervals; gain10 feasibility is explicit. Dashes mean the unconditional median ratio is undefined because most targets are infeasible.

Rule	Cap	Targets	Feas. (%)	L_1 ratio [95%]
Original board, all targets				
Mean	ceil	4495	100.0	1.000 [1.000, 1.000]
Minimum	ceil	4495	100.0	0.239 [0.237, 0.243]
Mean	gain10	4495	2.5	–
Minimum	gain10	4495	9.9	–
Original board, targets with no zero				
Mean	ceil	4179	100.0	1.000 [1.000, 1.000]
Minimum	ceil	4179	100.0	0.235 [0.233, 0.239]
Mean	gain10	4179	2.7	–
Minimum	gain10	4179	10.7	–
Original board, targets with a zero				
Mean	ceil	316	100.0	1.000 [1.000, 1.000]
Minimum	ceil	316	100.0	0.384 [0.379, 0.386]
Mean	gain10	316	0.0	–
Minimum	gain10	316	0.0	–
Zero-score models removed from board				
Mean	ceil	4179	100.0	1.000 [1.000, 1.000]
Minimum	ceil	4179	100.0	0.235 [0.233, 0.239]
Mean	gain10	4179	2.7	–
Minimum	gain10	4179	10.7	–

Table 32: Ordinal/mean per-target L_1 ratios at ceil. Brackets are 95% target-bootstrap intervals; the ratio statistic is a median of per-target ratios. Random-tail sample sizes differ because duplicate ranks collapse. All ceiling comparisons are feasible.

Rule	Targets	Median ratio [95%]	Cheaper / compared
Delivered grid			
Borda	136	0.458 [0.415, 0.505]	136/136
Plurality	136	0.416 [0.403, 0.455]	135/136
Condorcet req.	136	0.658 [0.648, 0.671]	134/136
Random-tail seed 20261108			
Borda	126	0.486 [0.449, 0.556]	126/126
Plurality	126	0.420 [0.381, 0.449]	125/126
Condorcet req.	126	0.655 [0.631, 0.668]	124/126
Random-tail seed 1			
Borda	130	0.490 [0.425, 0.549]	130/130
Plurality	130	0.411 [0.388, 0.445]	129/130
Condorcet req.	130	0.664 [0.656, 0.675]	128/130
Random-tail seed 2			
Borda	128	0.463 [0.432, 0.503]	128/128
Plurality	128	0.419 [0.398, 0.453]	127/128
Condorcet req.	128	0.657 [0.643, 0.676]	126/128

Table 33: Ordinal/mean per-target L_1 ratios at gain10. Brackets are 95% target-bootstrap intervals; the ratio statistic is a median of per-target ratios. Random-tail sample sizes differ because duplicate ranks collapse. Joint / total states the conditional cohort; the final column counts targets excluded because only the mean is feasible.

Rule	Joint / total	Median ratio [95%]	Mean feasible, ordinal infeasible
Delivered grid			
Borda	65/136	0.333 [0.281, 0.366]	2
Plurality	13/136	0.487 [0.357, 0.518]	54
Condorcet req.	43/136	0.546 [0.473, 0.583]	24
Random-tail seed 20261108			
Borda	52/126	0.325 [0.267, 0.369]	2
Plurality	13/126	0.487 [0.357, 0.518]	41
Condorcet req.	35/126	0.546 [0.469, 0.578]	19
Random-tail seed 1			
Borda	58/130	0.330 [0.275, 0.369]	3
Plurality	13/130	0.487 [0.357, 0.518]	48
Condorcet req.	36/130	0.546 [0.472, 0.583]	25
Random-tail seed 2			
Borda	55/128	0.333 [0.266, 0.371]	3
Plurality	14/128	0.478 [0.313, 0.511]	44
Condorcet req.	37/128	0.546 [0.469, 0.585]	21

Table 34: Observed boundary cases. front0 feasibility is complete; gain10 feasibility is not. Condorcet-winner existence under noise is counted over 50 OLLB and 200 HELM draws. Cost comparisons use mean nonleaders; existence uses the whole leaderboard.

Check	OLLB	HELM
Condorcet winner (observed)	No	Yes
Winner exists under noise	0/50	172/200
front0 feasible: mean / minimum	4495/4495 ; 4495/4495	78/78 ; 78/78
gain10 feasible: mean / minimum	113/4495 ; 446/4495	29/78 ; 14/78

E BOUNDARY CASES

AI USE STATEMENT

This work was carried out by generative-AI agents operating in separate roles: literature and context, study planning and feasibility, analysis code and experiments, figures and tables, venue and format, writing, and internal and external review. The writing role was Claude Sonnet 5.5 (Anthropic); the figures-and-tables role reports that it was Codex on GPT-6 (OpenAI); the agents of the other roles did not state their models to us and we do not attribute one to them. They were used for ideation of the study design, methodology design, implementation of the solvers and experiments, interpretation of results, and drafting of the text. No human author took part in these steps. Every number in the paper is produced by code in the repository and written into the text by a script; the solvers are tested against brute force on small random instances and the closed forms are compared with solver output; the delivered result files carry a manifest with checksums and an audit of completeness; and the roles reviewed each other's deliverables. Only references that the agents opened (title and abstract, or the full text where we say so) are cited, and the related work states only what those sources say. We take responsibility for the final content, including text, claims and artifacts produced with the aid of generative AI.